A Distributionally Robust Optimization Model for Vehicle Platooning under Stochastic Disturbances

Peiyu Zhang, Daxin Tian, *Senior Member, IEEE*, Jianshan Zhou, Xuting Duan, Dezong Zhao, *Senior Member, IEEE,* and Dongpu Cao, *Senior Member, IEEE,*

***Abstract*—Inspired by connected and autonomous driving technologies, this paper proposes a closed-loop Distributionally Robust Model Predictive Control (DRMPC) method to address the problem of longitudinal platoon control disturbed by V2V communication noise. In particular, a Model Predictive Control (MPC)-based vehicle platoon control model subject to stochastic disturbances is first developed. Vehicle control and state are imposed with probabilistic chance constraints, and a state feedback structure is designed to ensure the stability of the platoon system, which poses a significant challenge to the platoon control system. To solve this computationally intractable DRMPC model, a Ball ambiguity set is constructed using the characteristic information (expectation and variance) of random variables. The original DRMPC model is reformulated into a computationally tractable robust counterpart approximation framework. Furthermore, the recursive feasibility of the proposed DRMPC and the string stability of the platoon vehicles are demonstrated by introducing an initialization strategy for nominal states. Finally, a simulation study in a platooning system consisting of six vehicles is performed to verify the validity of the DRMPC model under stochastic V2V noise disturbances.**

***Index Terms*—Vehicle platooning, model predictive control, distributionally robust optimization, chance constraints, communication uncertainty.**

I. Introduction

In recent years, vehicle platoon control has attracted more and more attention because it is an effective way to improve traffic efficiency [1]–[4]. In a platooning system, each pair of consecutive vehicles enables information sharing among multiple vehicles in series through real-time vehicle-to-vehicles (V2V) communication information such as vehicle location, inter-vehicle distance, and inter-vehicle speed [5]. Real-time V2V data can develop a more robust and safer Cooperative Adaptive Cruise Control (CACC) system for vehicle platooning, enabling the two consecutive vehicles to maintain an ideal space headway and desired speed. Many academics have worked on developing different types of platooning control strategies to achieve this goal.

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Peiyu Zhang, Daxin Tian, Jianshan Zhou, Xuting Duan are with the Beijing Key Laboratory for Cooperative Vehicle Infrastructure Systems and Safety Control, Beijing Advanced Innovation Center for Big Data and Brain Computing, School of Transportation Science and Engineering, Beihang University, Beijing 100191, China (e-mail: zpeiyu2@163.com; jianshanzhou@foxmail.com; dtian@buaa.edu.cn; duanxuting@buaa.edu.cn).

Dezong Zhao is with the James Watt School of Engineering, University of Glasgow, Glasgow G12 8QQ, U.K. (e-mail: dezong.zhao@glasgow.ac.uk).

Dongpu Cao is with the School of Vehicle and Mobility, Tsinghua University, Beijing, 100084, China (e-mail: dongpu.cao@uwaterloo.ca).

However, traditional control methods for achieving safe and efficient platooning on the road still have challenges such as uncertainty in V2V communication caused by intercarrier interference. According to [6] and [7], a dynamic V2V channel environment generates random noise disturbances and these stochastic disturbances can lead to mismatches in the system model which consequently disrupts the performance of the platooning controller [8]. In this context, it is essential to develop some new control methods for vehicle platoon to guarantee that the following vehicles move steadily at the desired distance and speed even under the presence of stochastic noise. The objectives of this study are to develop a DRMPC modelling framework for vehicle platooning when V2V communication has stochastic noise and to improve driving safety, string stability, and robustness of a string of autonomous vehicles in an uncertain communication environment.

A. Related Work

In recent years, MPC has emerged as an advanced control method for developing platooning-oriented CACC problems. The MPC model solves the online control problems with a finite field of view and complex constraints at each sampling time along the prediction horizon. This results in obtaining a control sequence in the short-time domain. Such an approach holds promise for improved platooning performance in terms of safety and real-time availability [9]–[12]. In traditional MPC optimization problems, only the deterministic model, i.e., Nominal MPC (NMPC), is considered, which ignores the multiple uncertainties in the V2V communication process (time delays, position parameters, process noise). In this case, the NMPC controller may not be able to guarantee the safe operation of the platooning vehicles or meet the quality specifications [13], [14]. A Robust MPC (RMPC) can solve this class of vehicle formation problems with bounded disturbances if the bounds of the uncertain parameters can be known in advance [15], [16]. For example, [15] developed a new optimization method to solve the RMPC problem for linear discrete-time systems with security constraints and objectives in the presence of bounded disturbances. [16] designed a min-max robust controller for vehicle formation systems, the advantage of this controller is that it guarantees input-output stability when there exists a communication delay between vehicles. The design of the RMPC platoon controller is based on the worst-case scenario principle where the control system considers the maximum effect of uncertainty to ensure the feasibility of the safety constraints and the stability of the

system. As a result, the RMPC is usually too conservative and leads to a smaller feasible domain for the initial state, which can lead to no feasible solution to the optimization problem [17], [18].

To address the problems with RMPC, Stochastic MPC (SMPC) offers a new research idea [19]–[23]. If the probability distributions of uncertain parameters can be obtained in advance, then the information about these distributions (expectation, variance) can be applied to the construction of SMPC models. By introducing the chance constraint and specifying the maximum probability of violating this constraint, the conservatism inherent in the RMPC control method is improved. Therefore, a platooning control system using SMPC is an alternative method for dealing with uncertainty that can handle disturbances without fixed boundaries in a relatively large feasible domain [24], [25]. Moreover, [26] proposed the idea of a probability reachable set to tighten the chance constraint and realise the closed-loop stability under the single-mode assumption of the probability distribution. In general, both RMPC and SMPC are based on the setting of "perfect" information, i.e., the upper and lower bounds of uncertain parameters or the exact distribution of random parameters [27]. In practical applications, the availability of accurate real-time information or the exact distribution of random parameters is often limited, posing significant challenges for MPC-based platoon control with uncertain V2V communication [28].

As a new optimization method, distributionally robust optimization can solve SMPC models with an uncertain probability distribution. In the distributionally robust method, the realization of a random disturbance (probability distribution) belongs to a Ball convex set. This set contains all distributions that satisfy the same properties (e.g., expectation, variance and standard deviation), and they can be optimized for the worst-case scenario under the Ball convex set [29], [30]. There are several popular methods for constructing the convex sets, such as the Wasserstein distance [31]–[33], the likelihood constraint method [34] and the moment-based constraint method [29] (adopted here). In the present study, the moment-based DRO model, which uses characteristic information (mean and variance) to construct a ball ambiguity set, optimizes the expectation of the worst-case objective function under all distributions that satisfy the feature information.

To the best of our research, the currently available literature does not apply the "distributionally robust" to MPC platoon control under V2V communication with uncertainty. Existing studies of the MPC vehicle platoon control problem assume the known probability distributions given the bounded information in stochastic systems [8] or given uncertain parameters in min-max robust bounded system [35]. These approaches limit the improvement of the robustness and safety of platoon control systems in uncertain environments. Hence, it is essential to establish a DRMPC framework for platoon control under V2V communication with stochastic noise, which is capable of enhancing robustness against uncertainties and ensuring platoon stability.

B. Motivation and Contribution

This paper proposes a distributionally robust model-based predictive platoon control method to study the MPC-based vehicle platoon control problem in uncertain V2V communication conditions. The method obtains the optimal control sequence by optimizing the expected value of the objective function and a set of chance constraints with unknown distributions. Specifically, we impose probability chance constraints on both state and control, and design state feedback law to guarantee closed-loop stability of the platoon system. However, this increases the computational complexity of the DRMPC model. Drawing inspiration from [36], we present a robust counterpart reconstruction of the DRMPC model under a Ball convex set and conduct recursive feasibility and stability analyses. Furthermore, we perform a convergence analysis of the proposed DRMPC algorithm. To demonstrate the efficacy of our control method, we compare the performance of the DRMPC, NMPC, RMPC, and SMPC algorithms in uncertain scenarios for vehicle platoon control. We summarize the three significant contributions of this paper as follows:

- A novel DRMPC control approach is proposed for the platoon system when V2V communication has stochastic disturbances. Different from the control models with bounded disturbances and exact distributions, this study introduces a random disturbance with distributional characteristics to the platoon system, which is more practical.
- A DRMPC framework is developed to improve the robustness and stability of uncertain V2V communication data. Specifically, we construct a Ball ambiguity set based on the distribution information of random variables and derive a computationally tractable framework for DRMPC. In addition, a control algorithm is designed for the DRMPC model and the recursive feasibility and convergence are proved.
- The string stability of the proposed DRMPC-based model is demonstrated. Furthermore, we perform simulation experiments to verify the superior performance in terms of vehicle safety, robustness and stability by the proposed formation control method for predicting the driving state of controlled vehicles, compared to conventional MPC, and the interference-resistant RMPC and SMPC controllers.

The remainder of this paper is organized as follows. Section II develops the system model for a longitudinal platoon control problem under an uncertain V2V communication. In Section III, we deduce a robust counterpart approximation framework under the Ball ambiguity set and analyse the computational complexity, recursive feasibility and convergence for the proposed algorithm. Finally, Simulation studies are presented in Section IV. Conclusions and future research directions are presented in Section V.

II. PROBLEM FORMULATION AND SYSTEM MODEL

A. Vehicle Platoon System Model

The system scenario of the vehicle platoon and the control framework for the vehicle is shown in Figure 1. Specifically,

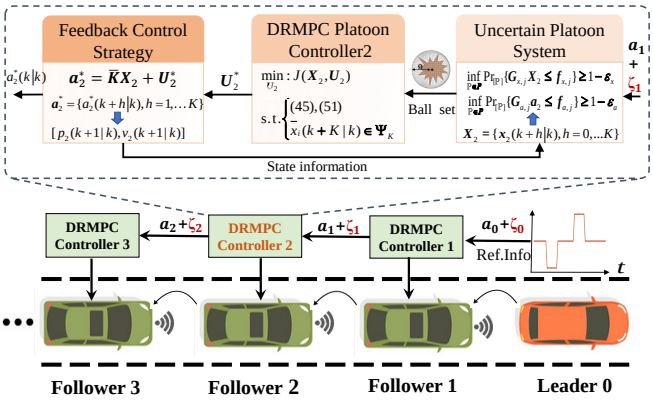


Fig. 1. The system scenario of vehicle platoon and the control framework of the following vehicle.

we consider a vehicle platoon on a straight road, which includes a leading vehicle (indexed as 0) with a known trajectory and its following vehicles. It can be observed that the front vehicle in the platoon can exchange information with the rear car via vehicular wireless communication, and an advanced controller installed in each following vehicle helps to manage the platoon's speed and distance between vehicles. To improve the robustness of the system, we propose the DRMPC controller to cope with the effect of random perturbations on the formation system. We purposefully construct the probabilistic chance constraints and objectives and obtain the robust approximation framework under the Ball convex set, and finally obtain the real-time control signals of the vehicles based on the feedback control strategy.

For simplicity, we define the set $\mathcal{I}_{N+1}$ to contain all platooning vehicles, i.e., $\mathcal{I}_{N+1} = \{0, 1, \ldots, N\}$, and the set $\mathcal{I}_N$ represents all following vehicles, i.e., $\mathcal{I}_N = \{\mathcal{I}_{N+1}\} - \{0\}$. The longitudinal dynamics model for vehicle i can be characterized by using the following form:

$$\begin{cases} p_i(k+1) = p_i(k) + \tau \cdot v_i(k) + \dfrac{\tau^2}{2} \cdot a_i(k) \\ v_i(k+1) = v_i(k) + \tau \cdot a_i(k), \quad i \in \mathcal{I}_N \end{cases} \tag{1}$$

where $p_i(k)$, $v_i(k)$ and $a_i(k)$ represent the longitudinal position, speed and acceleration of vehicle i at time k, respectively. τ is the sampling time, and $a_i(k)$ is control signal on each time intervel $[k\tau, (k+1)\tau]$ for $k \in \mathbb{Z}_+ := \{0, 1, 2, \ldots, K\}$. This double-integrator model contains the leading vehicle 0 and the followers' vehicles ($i \in \mathcal{I}_N$) which is widely applied to the system-level car-following control design problem.

For the leading vehicle, the trajectory of the vehicle is known in advance. Suppose vector $\triangle p_i(k)$ is the distance between vehicle $i-1$ and vehicle i, and $\triangle v_i(k)$ is the velocity difference between the two vehicles. Specifically, $\triangle p_i(k)$ and $\triangle v_i(k)$ have the following form

$$\begin{cases} \triangle\, p_i(k) = p_{i-1}(k) - p_i(k) - r \cdot v_i(k) - L, \\ \triangle\, v_i(k) = v_{i-1}(k) - v_i(k), \end{cases} \tag{2}$$

where $r \geq \tau$ is a constant representing the reaction time, and $L \geq 0$ is the vehicle length. Then, the dynamic system of the ith vehicle can be formulated as

$$\begin{aligned} \begin{bmatrix} \triangle p_i(k+1) \\ \triangle v_i(k+1) \end{bmatrix} &= \begin{bmatrix} 1 & \tau \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \triangle p_i(k) \\ \triangle v_i(k) \end{bmatrix} + \begin{bmatrix} \frac{\tau^2}{2} \\ \tau \end{bmatrix} a_{i-1}(k) \\ &+ \begin{bmatrix} -(\frac{\tau^2}{2} + r\tau) \\ -\tau \end{bmatrix} a_i(k), \end{aligned} \tag{3}$$

where $i \in \mathcal{I}_N$. Next, the system state is defined as $\boldsymbol{x}_i(k) = [\triangle p_i(k), \triangle v_i(k)]^{\mathrm{T}}$, and the dynamics system (3) can be represented as follows

$$\boldsymbol{x}_i(k+1) = \boldsymbol{A}\boldsymbol{x}_i(k) + \boldsymbol{B}a_i(k) + \boldsymbol{C}a_{i-1}(k) \tag{4}$$

where $\boldsymbol{A} = \begin{bmatrix} 1 & \tau \\ 0 & 1 \end{bmatrix}$, $\boldsymbol{B} = \begin{bmatrix} -(\frac{\tau^2}{2} + r\tau) \\ -\tau \end{bmatrix}$, $\boldsymbol{C} = \begin{bmatrix} \frac{\tau^2}{2} \\ \tau \end{bmatrix}$. In this system, the pair $(\boldsymbol{A}, \boldsymbol{B}, \boldsymbol{C})$ is stabilizable. During V2V communication between vehicle i and $i-1$, various types of noise and disturbances are often encountered, such as sensor measurement errors, communication delays, and model mismatch errors. Therefore, for the acceleration $a_{i-1}(k)$ of the preceding vehicle by V2V communication, we propose the following disturbance model:

$$a_{i-1}(k) = a^0_{i-1}(k) + \sum_{l=1}^{L} \zeta^l a^l_{i-1}, \tag{5}$$

where the true acceleration of vehicle $a_{i-1}(k)$ is composed of two parts, the nominal term $a^0_{i-1}(k)$ is the acceleration of vehicle $i-1$ at time k and the disturbance term $\sum_{l=1}^{L} \zeta^l a^l_{i-1}$. For the disturbance term, a^l_{i-1} is a constant disturbance term that represents the magnitude of the disturbance range, while ζ^l represents the random disturbance vector. Typically, we can obtain its expected and variance information, i.e., zero-mean and covariance matrix $\Sigma \succ 0$.

Remark 1: For each pair of consecutive vehicles, the vehicle in front only transmits its acceleration to the following vehicle through V2V communication. However, the high mobility of vehicles can cause a high Doppler frequency, which consequently leads to intercarrier interference [37]. Therefore, the communication information $a_{i-1}(k)$ is uncertain at time k. The assumption of $a_{i-1}(k)$ in (5) is a novel form by comparing with the other works.

B. Chance Constraints

Next, we summarise the basic state and control constraints. For vehicle i,

$$p_{\min} \leq \triangle p_i(k) - d_i \leq p_{\max}, \tag{6}$$

where $p_{\max}$ and $p_{\min}$ are the pre-specified upper bound and lower bound of the inter-vehicle distance error, respectively. d_i is the desired value of the inter-vehicle distance.

$$v_{\min} \leq \triangle v_i(k) \leq v_{\max}, \tag{7}$$

where $v_{\max} > 0$ and $v_{\min} \leq 0$ are the pre-specified upper bound and lower bound of the inter-vehicle speed error, respectively.

$$a_{\min} \leq a_i(k) \leq a_{\max}, \tag{8}$$

where $a_{\max} > 0$ and $a_{\min} < 0$ are the pre-specified minimum and the maximum accelerations for each vehicle. The constraints in (6) and (7) can improve the platooning performance. At the steady states, the inter-vehicle distance $\triangle p_i(k)$ of the platoon can be controlled to a fixed value d_i, while at transients, it is can take any value in the interval $[p_{\min} + d_i, p_{\max} + d_i]$. By the constraints (6) and (7), we conduct the state constraints of vehicle i as follows

$$\boldsymbol{x}_{\min} \leq \boldsymbol{x}_i(k) \leq \boldsymbol{x}_{\max}, \tag{9}$$

where $\boldsymbol{x}_{\min} = [p_{\min} + d_i, v_{\min}]^{\mathrm{T}}$ and $\boldsymbol{x}_{\max} = [p_{\max} + d_i, v_{\max}]^{\mathrm{T}}$. For simplicity, the state constraints (9) and control constraint (8) for improving the platoon performance can be represented by

$$\begin{cases} \boldsymbol{H}_x \boldsymbol{x}_i(k) \leq \boldsymbol{b}_x, i \in \mathcal{I}_N; & (10) \\ \boldsymbol{H}_a a_i(k) \leq \boldsymbol{b}_a, i \in \mathcal{I}_N, & (11) \end{cases}$$

where $\boldsymbol{H}_x = \mathrm{col}\{\boldsymbol{I}_{2\times 2}, -\boldsymbol{I}_{2\times 2}\}$, $\boldsymbol{b}_x = \mathrm{col}\{\boldsymbol{x}_{\max}, -\boldsymbol{x}_{\min}\}$, $\boldsymbol{H}_a = [1, -1]^{\mathrm{T}}$ and $\boldsymbol{b}_a = [a_{\max}, -a_{\min}]^{\mathrm{T}}$, respectively. In our setting, the matrix with respect to the bounds is fixed parameter matrix. Due to uncertain disturbances in the platoon system, we impose individual chance constraints on state and control of vehicle i

$$\begin{cases} \Pr_{[\mathbb{P}]}\{\boldsymbol{H}_{x,j}\boldsymbol{x}_i(k) \leq b_{x,j}\} \geq 1 - \epsilon_x, j = 1, \ldots, n_x; & (12) \\ \Pr_{[\mathbb{P}]}\{\boldsymbol{H}_{a,j} a_i(k) \leq b_{a,j}\} \geq 1 - \epsilon_a, j = 1, \ldots, n_a, & (13) \end{cases}$$

where $\boldsymbol{H}_{x,j}$ is row j of the matrix $\boldsymbol{H}_x$, $\boldsymbol{H}_{a,j}$ is row j of the matrix $\boldsymbol{H}_a$, $b_{x,j}$ is the jth element in $\boldsymbol{b}_x$ and $b_{a,j}$ is the jth element in $\boldsymbol{b}_a$. n_x and n_a are the row indexes to the matrices $\boldsymbol{H}_{x,j}$ and $\boldsymbol{H}_{a,j}$. Besides, ϵ_x and $\epsilon_a \in (0, 1)$ are violation values of the chance constraints. $\mathbb{P}$ is the probability distribution of disturbance vector ζ.

Remark 2: In real applications, the constraints on the control input $a_i(k)$ are generally hard, i.e., in the form of constraint (8). However, the control input is influenced by the disturbance vector ζ if a state feedback control strategy is imposed. If ζ is unbounded and only the partial probability distribution information (mean and variance) is known, then it is difficult to enforce hard constraints on the control input $a_i(k)$ and solve them. In this paper, a Ball convex set is defined for the unbounded random disturbance and we deduce a computationally tractable framework for the chance constraints.

To proceed further, for each following vehicle $i \in \mathcal{I}_N$ at every time k, we define the probability set $\mathcal{P}_k$ as follow

$$\mathcal{P}_k = \left\{\mathbb{P} : \mathbb{E}_{[\mathbb{P}]}\left\{\zeta^l\right\} = 0, \mathbb{E}_{[\mathbb{P}]}\left\{(\zeta^l)(\zeta^l)^{\mathrm{T}}\right\} = \Sigma\right\}. \tag{14}$$

In probability set $\mathcal{P}_k$, we only know the partial probability information which makes $\mathbb{P} \in \mathcal{P}_k$ not a fixed distribution. Based on the "worst case" principle, we impose a distributionally robust version of chance constraints (12) and (13) as

$$\inf_{\mathbb{P}\in\mathcal{P}_k} \Pr_{[\mathbb{P}]}\{\boldsymbol{H}_{x,j}\boldsymbol{x}_i(k) \leq b_{x,j}\} \geq 1 - \epsilon_x, j = 1, \ldots, n_x \tag{15}$$

and

$$\inf_{\mathbb{P}\in\mathcal{P}_k} \Pr_{[\mathbb{P}]}\{H_{a,j} a_i(k) \leq b_{a,j}\} \geq 1 - \epsilon_a, j = 1, \ldots, n_a. \tag{16}$$

Remark 3: According to [38], [39], the specific meaning of the term "distributionally robust" means that it is worst-case robust among all probability distributions that satisfy the distribution characteristics (mean, variance), and we optimize the worse-case distribution $\mathbb{P}$ that belongs to an ambiguity set $\mathcal{P}_k$.

C. Feedback Control Law

In order to achieve a closed-loop effect for each DRMPC controller, the following linear state feedback strategy is used:

$$a_i(k) = \boldsymbol{K}\boldsymbol{x}_i(k) + u_i(k), i \in \mathcal{I}_N, \tag{17}$$

where gain matrix $\boldsymbol{K}$ is determined off-line such that dynamics system (4) is stable with $a_i(k) = \boldsymbol{K}\boldsymbol{x}_i(k)$. According to the literature [40] and [41], this strategy parameterise the control signals by linearly combining the system's state $\boldsymbol{x}_i(k)$, and the control gain $\boldsymbol{K}$ can be computed by a LQR optimization problem [40], and the $u_i(k)$ is a new decision variable that uniquely determines the control input $a_i(k)$. Moreover, the LQR-based Riccati equation is also important to ensure the convergence of the model.

In addition, we use the idea of rolling optimization to solve the model. For each following vehicle $i \in \mathcal{I}_N$ at time k, substituting (17) into (4), we can obtain the following state equation

$$\boldsymbol{x}_i(k+1) = (\boldsymbol{A} + \boldsymbol{B}\boldsymbol{K})\boldsymbol{x}_i(k) + \boldsymbol{B}u_i(k) + \boldsymbol{C}a_{i-1}(k). \tag{18}$$

In this paper, we employ the MPC method for the rolling optimization of vehicular control signals. In order to derive a more compact form over the prediction horizon, we assume that k is the initial instant, K is the terminal predicted instant, $\boldsymbol{x}_i(k+h|k)$ and $u_i(k+h|k)$ are the predicted state and predicted control signal, among that $h = 1, ..., K$ and $k < K$. Besides, $\boldsymbol{x}_i(k|k)$ is a known initial state of the platoon system. Thus, we denote

$$\begin{cases} \boldsymbol{X}_i = \mathrm{col}\{\boldsymbol{x}_i(k+h|k), h = 1, ..., K\}, \\ \boldsymbol{U}_i = \mathrm{col}\{u_i(k+h|k), h = 0, 1, ..., K-1\}, \\ \boldsymbol{D}_{i-1} = \mathrm{col}\{a_{i-1}(k+h|k), h = 0, 1, ..., K-1\}, \\ \boldsymbol{\varphi} = \boldsymbol{A} + \boldsymbol{B}\boldsymbol{K}. \end{cases} \tag{19}$$

The system state equation (18) of all prediction horizons can be compactly reshaped as

$$\boldsymbol{X}_i = \boldsymbol{M}_{\boldsymbol{\varphi}}\boldsymbol{x}_i(k|k) + \boldsymbol{M}_B\boldsymbol{U}_i + \boldsymbol{M}_C\boldsymbol{D}_{i-1}, \tag{20}$$

where the system matrices $\boldsymbol{M}_{\boldsymbol{\varphi}}$, $\boldsymbol{M}_B$ and $\boldsymbol{M}_C$ are defined by

$$\boldsymbol{M}_{\boldsymbol{\varphi}} = \mathrm{diag}\left\{\boldsymbol{\varphi}^1, \boldsymbol{\varphi}^2, \ldots, \boldsymbol{\varphi}^K\right\},$$

$$\boldsymbol{M}_B = \begin{bmatrix} \boldsymbol{B} & \boldsymbol{0} & \boldsymbol{0} & \cdots & \boldsymbol{0} & \boldsymbol{0} \\ \boldsymbol{\varphi}\boldsymbol{B} & \boldsymbol{B} & \boldsymbol{0} & \cdots & \boldsymbol{0} & \boldsymbol{0} \\ \boldsymbol{\varphi}^2\boldsymbol{B} & \boldsymbol{\varphi}\boldsymbol{B} & \boldsymbol{B} & \cdots & \boldsymbol{0} & \boldsymbol{0} \\ \vdots & \vdots & \vdots & \cdots & \vdots & \vdots \\ \boldsymbol{\varphi}^{K-1}\boldsymbol{B} & \boldsymbol{\varphi}^{K-2}\boldsymbol{B} & \boldsymbol{\varphi}^{K-3}\boldsymbol{B} & \cdots & \boldsymbol{\varphi}\boldsymbol{B} & \boldsymbol{B} \end{bmatrix},$$

$$\boldsymbol{M}_C = \begin{bmatrix} \boldsymbol{C} & \boldsymbol{0} & \boldsymbol{0} & \cdots & \boldsymbol{0} & \boldsymbol{0} \\ \boldsymbol{\varphi}\boldsymbol{C} & \boldsymbol{C} & \boldsymbol{0} & \cdots & \boldsymbol{0} & \boldsymbol{0} \\ \boldsymbol{\varphi}^2\boldsymbol{C} & \boldsymbol{\varphi}\boldsymbol{C} & \boldsymbol{C} & \cdots & \boldsymbol{0} & \boldsymbol{0} \\ \vdots & \vdots & \vdots & \ldots & \vdots & \vdots \\ \boldsymbol{\varphi}^{K-1}\boldsymbol{C} & \boldsymbol{\varphi}^{K-2}\boldsymbol{C} & \boldsymbol{\varphi}^{K-3}\boldsymbol{C} & \cdots & \boldsymbol{\varphi}\boldsymbol{C} & \boldsymbol{C} \end{bmatrix}.$$

Due to the V2V communication signals $\boldsymbol{D}_{i-1}$ is uncertain, and has the following disturbance form

$$\begin{aligned}\boldsymbol{D}_{i-1} &= \boldsymbol{D}_{i-1}^{0} + \boldsymbol{\zeta}\boldsymbol{D}_{i-1}^{\zeta} \\ &= \begin{bmatrix} a_{i-1}^{0}(k|k) \\ a_{i-1}^{0}(k+1|k) \\ a_{i-1}^{0}(k+2|k) \\ \vdots \\ a_{i-1}^{0}(k+K-1|k) \end{bmatrix} + \begin{bmatrix} \zeta^1 & \zeta^2 & \cdots & \zeta^L \\ \zeta^1 & \zeta^2 & \cdots & \zeta^L \\ \zeta^1 & \zeta^2 & \cdots & \zeta^L \\ \vdots & \vdots & \cdots & \vdots \\ \zeta^1 & \zeta^2 & \cdots & \zeta^L \end{bmatrix} \begin{bmatrix} a_{i-1}^{1} \\ a_{i-1}^{2} \\ a_{i-1}^{3} \\ \vdots \\ a_{i-1}^{L} \end{bmatrix},\end{aligned} \tag{21}$$

where the nominal matrix $\boldsymbol{D}_{i-1}^{0}$, disturbance matrix $\boldsymbol{D}_{i-1}^{\zeta}$ are defined by

$$\begin{aligned}\boldsymbol{D}_{i-1}^{0} &= \mathrm{col}\{a_{i-1}^{0}(k+h|k), h = 0, 1, ..., K-1\}, \\ \boldsymbol{D}_{i-1}^{\zeta} &= \mathrm{col}\{a_{i-1}^{l}, l = 1, ..., L\}.\end{aligned}$$

In addition, $\boldsymbol{\zeta}$ is an uncertain variable with stochastic information and a_{i-1}^{l} is a constant which represents the size of the perturbation. Up to now, we can obtain the final system state equation of vehicle i in all prediction horizon

$$\boldsymbol{X}_i = \boldsymbol{M}_{\boldsymbol{\varphi}}\boldsymbol{x}_i(k|k) + \boldsymbol{M}_B\boldsymbol{U}_i + \boldsymbol{M}_C(\boldsymbol{D}_{i-1}^{0} + \boldsymbol{\zeta}\boldsymbol{D}_{i-1}^{\zeta}). \tag{22}$$

And the feedback control strategy (17) can be rewritten as

$$\begin{aligned}\boldsymbol{a}_i &= \bar{\boldsymbol{K}}\boldsymbol{X}_i + \boldsymbol{U}_i \\ &= \bar{\boldsymbol{K}}\boldsymbol{M}_{\boldsymbol{\varphi}}\boldsymbol{x}_i(k|k) + \bar{\boldsymbol{K}}\boldsymbol{M}_B\boldsymbol{U}_i + \boldsymbol{U}_i + \bar{\boldsymbol{K}}\boldsymbol{M}_C(\boldsymbol{D}_{i-1}^{0} + \boldsymbol{\zeta}\boldsymbol{D}_{i-1}^{\zeta}),\end{aligned} \tag{23}$$

where $\bar{\boldsymbol{K}} = \mathrm{diag}\{\boldsymbol{K}, h = 1, \ldots, K\}$. Further, the compact form of the state chance constrain (15) over all prediction horizons can be rewritten as

$$\inf_{\mathbb{P}\in\mathcal{P}} \Pr_{[\mathbb{P}]}\{\boldsymbol{G}_{x,j}\boldsymbol{X}_i \le f_{x,j}\} \ge 1 - \epsilon_x, j = 1, \ldots, n_x, \tag{24}$$

where $\boldsymbol{G}_x = \mathrm{diag}\{\boldsymbol{H}_x, h = 1, \ldots, K\}$, $\boldsymbol{f}_x = \mathrm{diag}\{\boldsymbol{b}_x, h = 1, \ldots, K\}$. j is the row index of all matrices and n_x is the row index to the matrices $\boldsymbol{G}_x$. Similarly, we can deduce the compact form of the control chance constrain (16) over all prediction horizons:

$$\inf_{\mathbb{P}\in\mathcal{P}} \Pr_{[\mathbb{P}]}\{\boldsymbol{G}_{a,j}\boldsymbol{a}_i \le f_{a,j}\} \ge 1 - \epsilon_a, j = 1, \ldots, n_a, \tag{25}$$

where n_a is the row index to the matrices $\boldsymbol{f}_x$. Constraints (24) and (25) are based on the worst case to satisfy the probability constraint. Constraint (24) enforces that the state of vehicle i should satisfy the inequalities $\boldsymbol{G}_{x,j}\boldsymbol{X}_i \le f_{x,j}$ with lowest probability $1 - \epsilon_x$ and $\epsilon_x \in (0, 1)$. Similarly, Constraint (25) indicates that the control signal should satisfy the inequalities $\boldsymbol{G}_{a,j}\boldsymbol{a}_i \le f_{a,j}$ with lowest probability $1 - \epsilon_a$ and $\epsilon_a \in (0, 1)$. The corresponding ambiguity set has the following form:

$$\mathcal{P} = \left\{\mathbb{P} : \mathbb{E}_{[\mathbb{P}]}\{\boldsymbol{\zeta}\} = \boldsymbol{0}, \mathbb{E}_{[\mathbb{P}]}\left\{\boldsymbol{\zeta}\boldsymbol{\zeta}^{\mathrm{T}}\right\} = \boldsymbol{\Sigma}\right\} \tag{26}$$

is obtained straight forward from (14). Each element of the random variable matrix $\boldsymbol{\zeta}$ is independent and $\boldsymbol{0} = \boldsymbol{1}_{K\times L} \otimes 0$, $\boldsymbol{\Sigma} = \boldsymbol{1}_{K\times L} \otimes \Sigma$, $\boldsymbol{1}_{K\times L}$ denotes an $K \times L$ with all elements being 1.

D. Objective Function and Terminal Constraints

In this paper, the control objective of DRMPC design for platooning vehicle i is to achieve offset cancellation of the inter-vehicle state and reduce the control effort. We adopt the most used quadratic function in a stochastic version as the objective function:

$$\begin{aligned}J(\boldsymbol{X}_i, \boldsymbol{U}_i) = \mathbb{E}_{[\mathbb{P}]}\Bigg[&\sum_{h=0}^{K-1} f(\boldsymbol{x}_i(k+h|k), a_i(k+h|k)) \\ &+ \Phi(\boldsymbol{x}_i(k+K|k))\Bigg],\end{aligned} \tag{27}$$

where $f(\boldsymbol{x}_i(k+h|k), a_i(k+h|k))$ is the stage cost function which is expressed as

$$\begin{aligned}&f(\boldsymbol{x}_i(k+h|k), a_i(k+h|k)) \\ &= \boldsymbol{x}_i^{\mathrm{T}}(k+h|k)\boldsymbol{Q}x_i(k+h|k) + a_i^{\mathrm{T}}(k+h|k)Ra_i(k+h|k),\end{aligned} \tag{28}$$

$\Phi(\boldsymbol{x}_i(k+K|k))$ is the terminal function and its expression is

$$\Phi(\boldsymbol{x}_i(k+K|k)) = \boldsymbol{x}_i^{\mathrm{T}}(k+K|k)\boldsymbol{Q}_K x_i(k+K|k) \tag{29}$$

In addition, $\boldsymbol{Q} = \begin{bmatrix} Q_1 & 0 \\ 0 & Q_2 \end{bmatrix}$ is a weight matrix which is positive definite and symmetric, $R > 0$ and $\boldsymbol{Q}_K$ can be solved by the following Riccati equation

$$(\boldsymbol{A} + \boldsymbol{B}\boldsymbol{K})\boldsymbol{Q}_K(\boldsymbol{A} + \boldsymbol{B}\boldsymbol{K}) - \boldsymbol{Q}_K + \boldsymbol{Q} + \boldsymbol{K}^{\mathrm{T}}R\boldsymbol{K} = 0. \tag{30}$$

Based on the probability distribution information of the random variable, we define

$$\bar{\boldsymbol{x}}_i(k+h|k) = \mathbb{E}_{[\mathbb{P}]}\left\{\boldsymbol{x}_i(k+h|k)\right\}, h = 0, 1, \ldots, K, \tag{31}$$

where $\mathbb{P} \in \mathcal{P}$. Due to the $\mathbb{E}_{[\mathbb{P}]}\{\boldsymbol{\zeta}\} = \boldsymbol{0}$, the expectation of the state equation (18) is

$$\bar{\boldsymbol{x}}_i(k+1|k) = \boldsymbol{\varphi}\bar{\boldsymbol{x}}_i(k|k) + \boldsymbol{B}u_i(k|k) + \boldsymbol{C}a_{i-1}^{0}(k|k), \tag{32}$$

and the compact form of system state equation (18) has the following form

$$\bar{\boldsymbol{X}}_i = \boldsymbol{M}_{\boldsymbol{\varphi}}\bar{\boldsymbol{x}}_i(k|k) + \boldsymbol{M}_B\boldsymbol{U}_i + \boldsymbol{M}_C\boldsymbol{D}_{i-1}^{0}. \tag{33}$$

To guarantee the stability of the system, we impose the following terminal constraints, which consist of nominal states

$$\bar{\boldsymbol{x}}_i(k+K|k) \in \Theta_K. \tag{34}$$

where the positively invariant set Θ_K satisfies

$$\boldsymbol{\varphi}\bar{\boldsymbol{x}}_i(k+K|k) \in \Theta_K, \forall\bar{\boldsymbol{x}}_i(k+K|k) \in \Theta_K. \tag{35}$$

E. Optimization Problem

In this subsection, we formally propose the DRMPC problem for the controller in platooning vehicle i. Under the

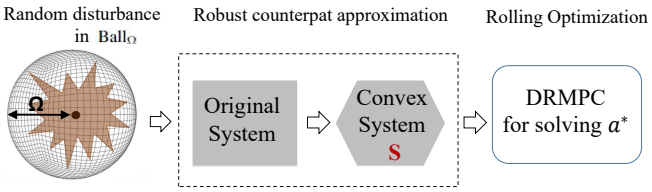


Fig. 2. The reformulation process under Ball$_\Omega$.

known initial state $\boldsymbol{x}_i(k|k)$ at time k, we propose the following optimization model **DRMPC-M1**:

$$\begin{aligned}
\min_{\boldsymbol{U}_i}\ & J(\boldsymbol{X}_i,\boldsymbol{U}_i)\\
\text{s.t.}\ & \boldsymbol{X}_i=\boldsymbol{M}_{\boldsymbol{\varphi}}\boldsymbol{x}_i(k|k)+\boldsymbol{M}_B\boldsymbol{U}_i+\boldsymbol{M}_C(\boldsymbol{D}_{i-1}^0+\boldsymbol{\zeta}\boldsymbol{D}_{i-1}^{\zeta}),\\
& \bar{\boldsymbol{X}}_i=\boldsymbol{M}_{\boldsymbol{\varphi}}\bar{\boldsymbol{x}}_i(k|k)+\boldsymbol{M}_B\boldsymbol{U}_i+\boldsymbol{M}_C\boldsymbol{D}_{i-1}^0,\\
& \boldsymbol{a}_i=\bar{\boldsymbol{K}}\boldsymbol{X}_i+\boldsymbol{U}_i,\\
& \inf_{\mathbb{P}\in\mathcal{P}}\Pr_{[\mathbb{P}]}\{\boldsymbol{G}_{x,j}\boldsymbol{X}_i\le f_{x,j}\}\ge 1-\epsilon_x, j=1,\dots,n_x,\\
& \inf_{\mathbb{P}\in\mathcal{P}}\Pr_{[\mathbb{P}]}\{\boldsymbol{G}_{a,j}\boldsymbol{a}_i\le f_{a,j}\}\ge 1-\epsilon_a, j=1,\dots,n_a,\\
& \bar{\boldsymbol{x}}_i(k+K|k)\in\Theta_K,\ i\in\mathcal{I}_N.
\end{aligned}\tag{36}$$

The non-convex optimization model **DRMPC-M1** is difficult or even impossible to solve because of the ambiguous chance constraints. In addition, the quadratic objective function also brings challenges to the solution of the model. The following section dresses the ambiguous chance constraints and the objective function (27) in **DRMPC-M1** based on the robust counterpart approximation theory, which can reformulate the model **DRMPC-M2** into a computationally tractable approximation framework.

III. Tractable Reformulation

Through the discussion of the non-convex model **DRMPC-M1**, it can be observed that it is computationally intractable due to the expected objective and probability chance constraints (24) and (25). To address such problems, we employ the distributionally robust counterpart theory to derive an approximate framework that is computationally tractable for model **DRMPC-M2**. Specifically, we derive the conic representation of the expected objective function (27), which is dependent on the decision variables. Then, we obtain two safe convex approximations of chance constraints (24) and (25) under a Ball convex set. These theoretical results provide a practical and effective approach to handling the computational challenges of **DRMPC-M1**.

As shown in Figure 2, we provide the process of reformulating model **DRMPC-M1** by using a Ball convex set. Building on the work of [39], we demonstrate the existence of a convex approximation system **S** under the Ball set of radius Ω. Utilizing this system, we develop a robust counterpart framework of **DRMPC-M1** which is computationally tractable. Notably, this is the first attempt to analyze the theory under the Ball ambiguity set, which is different from the robust min-max control method for bounded perturbations (box uncertainty sets). The model under this set can resist stochastic disturbances and prevent overly conservative intervention measures, thus avoiding excessive costs for decision-makers.

A. Tractable Reformulation of the Objective Function

Theorem 1: When the every element of the random matrix $\boldsymbol{\zeta}$ is independent and $\boldsymbol{\zeta}$ obeys the probability distribution $\mathbb{P}$, the objective function (27) is equivalent to

$$J(\boldsymbol{X}_i,\boldsymbol{U}_i)=U_i^{\mathrm{T}}\Theta U_i+\mathbf{h}^{\mathrm{T}}U_i+s \tag{37}$$

where

$$\begin{aligned}
s=&\bar{\boldsymbol{x}}_i^{\mathrm{T}}(k|k)\boldsymbol{M}_{\boldsymbol{\varphi}}^{\mathrm{T}}\boldsymbol{Q}_x\boldsymbol{M}_{\boldsymbol{\varphi}}\bar{\boldsymbol{x}}_i(k|k)+2\bar{\boldsymbol{x}}_i^{\mathrm{T}}(k|k)\boldsymbol{M}_{\boldsymbol{\varphi}}^{\mathrm{T}}\boldsymbol{Q}_x\boldsymbol{M}_C\boldsymbol{D}_{i-1}^0\\
&+(\boldsymbol{D}_{i-1}^0)^{\mathrm{T}}\boldsymbol{M}_C^{\mathrm{T}}\boldsymbol{Q}_x\boldsymbol{M}_C\boldsymbol{D}_{i-1}^0\\
&+(\boldsymbol{D}_{i-1}^{\zeta})^{\mathrm{T}}\mathrm{tr}\left\{\boldsymbol{M}_C^{\mathrm{T}}\boldsymbol{Q}_x\boldsymbol{M}_C\boldsymbol{\Sigma}\right\}\boldsymbol{D}_{i-1}^{\zeta}\\
&+\bar{\boldsymbol{x}}_i^{\mathrm{T}}(k|k)\boldsymbol{M}_{\boldsymbol{\varphi}}^{\mathrm{T}}\bar{\boldsymbol{K}}^{\mathrm{T}}\boldsymbol{R}_u\bar{\boldsymbol{K}}\boldsymbol{M}_{\boldsymbol{\varphi}}\bar{\boldsymbol{x}}_i(k|k)\\
&+2(\boldsymbol{D}_{i-1}^0)^{\mathrm{T}}\boldsymbol{M}_C^{\mathrm{T}}\bar{\boldsymbol{K}}^{\mathrm{T}}\boldsymbol{R}_u\bar{\boldsymbol{K}}\boldsymbol{M}_{\boldsymbol{\varphi}}\bar{\boldsymbol{x}}_i(k|k)\\
&+(\boldsymbol{D}_{i-1}^0)^{\mathrm{T}}\boldsymbol{M}_C^{\mathrm{T}}\bar{\boldsymbol{K}}^{\mathrm{T}}\boldsymbol{R}_u\bar{\boldsymbol{K}}\boldsymbol{M}_C\boldsymbol{D}_{i-1}^0\\
&+(\boldsymbol{D}_{i-1}^{\zeta})^{\mathrm{T}}\mathrm{tr}\left\{\boldsymbol{M}_C^{\mathrm{T}}\bar{\boldsymbol{K}}^{\mathrm{T}}\boldsymbol{R}_u\bar{\boldsymbol{K}}\boldsymbol{M}_C\boldsymbol{\Sigma}\right\}\boldsymbol{D}_{i-1}^{\zeta}.
\end{aligned}\tag{38}$$

$$\Theta=\boldsymbol{M}_B^{\mathrm{T}}\boldsymbol{Q}_x\boldsymbol{M}_B+(\bar{\boldsymbol{K}}\boldsymbol{M}_B+\boldsymbol{I})^{\mathrm{T}}\boldsymbol{R}_u(\bar{\boldsymbol{K}}\boldsymbol{M}_B+\boldsymbol{I}), \tag{39}$$

$$\begin{aligned}
\mathbf{h}^{\mathrm{T}}=&2\bar{\boldsymbol{x}}_i^{\mathrm{T}}(k|k)\boldsymbol{M}_{\boldsymbol{\varphi}}^{\mathrm{T}}\boldsymbol{Q}_x\boldsymbol{M}_B+2\left(\boldsymbol{D}_{i-1}^0\right)^{\mathrm{T}}\boldsymbol{M}_C^{\mathrm{T}}\boldsymbol{Q}_x\boldsymbol{M}_B\\
&+2\bar{\boldsymbol{x}}_i^{\mathrm{T}}(k|k)\boldsymbol{M}_{\boldsymbol{\varphi}}^{\mathrm{T}}\bar{\boldsymbol{K}}^{\mathrm{T}}\boldsymbol{R}_u(\bar{\boldsymbol{K}}\boldsymbol{M}_B+\boldsymbol{I})\\
&+2(\boldsymbol{D}_{i-1}^0)^{\mathrm{T}}\boldsymbol{M}_C^{\mathrm{T}}\bar{\boldsymbol{K}}^{\mathrm{T}}\boldsymbol{R}_u(\bar{\boldsymbol{K}}\boldsymbol{M}_B+\boldsymbol{I}),
\end{aligned}\tag{40}$$

proof: Please refer to Appendix A. □

B. Convex Approximations of Chance Constraints

In general, chance constraints (24) and (25) with unknown probability distributions are computationally intractable. However, in a distributionally robust framework, we can use a convex system under the Ball ambiguity set to approximate these constraints. According to [38], we first derive the robust counterpart frameworks of the linear inequality constraints in the chance constraints (24) and (25) under the Ball ambiguity set, then we shall prove that the robust counterparts are the safe convex approximations of the constraints.

In the case of the distribution information (26), the random matrix $\boldsymbol{\zeta}$ belongs to the following Ball convex set is defined.

$$\text{Ball}_\Omega=\{\zeta_k^l\in\boldsymbol{\zeta}:\|\zeta_k^l/\Sigma\|_2\le\Omega\}, \tag{41}$$

where Ω is the radius of ball. The details of robust counterpart approximations under this Ball set (41) are summarized into the following Lemmas and Theorems.

Lemma 1 (The Robust Counterpart [38]): Consider the random variable ζ belongs to Ball ambiguity set, the linear inequality constraint

$$Cx\le d,\forall\left(C=C_0+\sum_{l=1}^{L}\zeta_l C_l:\zeta\in\text{Ball}_\Omega\right) \tag{42}$$

has the following robust counterpart framework

$$\Omega\sqrt{\sum_{l=1}^{L}(C_l x)^2}\le d-C_0x. \tag{43}$$

where C is uncertain parameter, d is a real-valued parameter and x is decision variable.

Lemma 2 (Proposition 2.3.1 in [38]): Let $C^l, l = 1, \ldots, L$ are the disturbance parameters and $\zeta_l, l = 1, \ldots, L$ are independent random variables with $\mathbb{E}\{\zeta_l\} = 0$ and $\Sigma = 1$ taking values in $[-1, 1]$. Then for every $\Omega \geq 0$, it have

$$\Pr\left\{\zeta : \sum_{l=1}^{L} \zeta_l C_l > \Omega \sqrt{\sum_{l=1}^{L} C_l^2}\right\} \leq \exp\left\{-\Omega^2/2\right\}. \quad (44)$$

Theorem 2: For each DRMPC controller of vehicle i, the state chance constraints (24) has the following robust counterpart approximations framework under Ball ambiguity set:

$$\begin{cases} \boldsymbol{G}_x(\boldsymbol{M}_{\boldsymbol{\varphi}}\bar{\boldsymbol{x}}_i(k|k) + \boldsymbol{M}_B\boldsymbol{U}_i) \leq \boldsymbol{f}_x \\ -\boldsymbol{M}_C\boldsymbol{D}_{i-1}^0 - \boldsymbol{M}_C\boldsymbol{\Omega}_2\sqrt{(\boldsymbol{D}_{i-1}^{\zeta})^2}; \\ \boldsymbol{\Omega}_1 \geq \sqrt{2\ln(1/\epsilon_x)} \otimes \mathbf{1}_{K\times K}. \end{cases} \quad (45)$$

proof: The body of the chance constraint (24) is

$$\boldsymbol{G}_x\boldsymbol{X}_i \leq \boldsymbol{f}_x. \quad (46)$$

Substituting system state equation (22) in the linear inequality (46), we have

$$\boldsymbol{G}_x\left(\boldsymbol{M}_{\boldsymbol{\varphi}}\bar{\boldsymbol{x}}_i(k|k)\boldsymbol{M}_B\boldsymbol{U}_i + \boldsymbol{M}_C(\boldsymbol{D}_{i-1}^0 + \boldsymbol{\zeta}\boldsymbol{D}_{i-1}^{\zeta})\right) \leq \boldsymbol{f}_x. \quad (47)$$

Applying $\boldsymbol{\zeta} \in \text{Ball}_\Omega$ and Lemma 1, we have

$$\begin{aligned} \max_{\|\boldsymbol{\zeta}\|_2 \leq \boldsymbol{\Omega}_1} \boldsymbol{G}_x\boldsymbol{M}_C(\boldsymbol{\zeta}\boldsymbol{D}_{i-1}^{\zeta}) \leq& \boldsymbol{f}_x - \boldsymbol{G}_x\left(\boldsymbol{M}_{\boldsymbol{\varphi}}\bar{\boldsymbol{x}}_i(k|k) + \boldsymbol{M}_B\boldsymbol{U}_i\right) \\ &- \boldsymbol{G}_x\boldsymbol{M}_C\boldsymbol{D}_{i-1}^0 \\ \Rightarrow \boldsymbol{G}_x\boldsymbol{M}_C\boldsymbol{\Omega}_1\sqrt{(\boldsymbol{D}_{i-1}^{\zeta})^2} \leq& \boldsymbol{f}_x - \boldsymbol{G}_x\left(\boldsymbol{M}_{\boldsymbol{\varphi}}\bar{\boldsymbol{x}}_i(k|k) + \boldsymbol{M}_B\boldsymbol{U}_i\right) \\ &- \boldsymbol{G}_x\boldsymbol{M}_C\boldsymbol{D}_{i-1}^0. \end{aligned} \quad (48)$$

where $\boldsymbol{\Omega}_1 = \sqrt{2\ln(1/\epsilon_x)} \otimes \mathbf{1}_{K\times K}$.

Next, we shall prove that if (48) is the robust counterpart framework of the linear inequality $\boldsymbol{G}_x\boldsymbol{X}_i \leq \boldsymbol{f}_x$ and (26) takes place, then the decision variable $\boldsymbol{U}_i$ is feasible for (48) with probability at least $1 - \epsilon_x$. When $\|\zeta_k^l\|_2 \leq \Omega$, we have

$$\begin{aligned} &\boldsymbol{G}_x\boldsymbol{M}_C\boldsymbol{\zeta}\boldsymbol{D}_{i-1}^{\zeta} > \boldsymbol{f}_x - \boldsymbol{G}_x\left(\boldsymbol{M}_{\boldsymbol{\varphi}}\bar{\boldsymbol{x}}_i(k|k) + \boldsymbol{M}_B\boldsymbol{U}_i\right) - \boldsymbol{G}_x\boldsymbol{M}_C\boldsymbol{D}_{i-1}^0 \\ &\Rightarrow \boldsymbol{G}_x\boldsymbol{M}_C\boldsymbol{\zeta}\boldsymbol{D}_{i-1}^{\zeta} > \boldsymbol{G}_x\boldsymbol{M}_C\boldsymbol{\Omega}_1\sqrt{(\boldsymbol{D}_{i-1}^{\zeta})^2} \quad \text{[by (48)]} \\ &\Rightarrow \boldsymbol{\zeta}\boldsymbol{D}_{i-1}^{\zeta} > \boldsymbol{\Omega}_1\sqrt{(\boldsymbol{D}_{i-1}^{\zeta})^2}. \end{aligned}$$

By Lemma 2, we can obtain

$$\Pr\left\{\boldsymbol{\zeta}_j \in \boldsymbol{\zeta} : \boldsymbol{\zeta}_j\boldsymbol{D}_{i-1}^{\zeta} > \Omega_1\sqrt{(D_{i-1,j}^{\zeta})^2}\right\} \leq \exp\left\{-\Omega_1^2/2\right\}, \quad (49)$$

where $j = 1, \ldots, n_D$. Similarly, n_D is the row indexes to the matrices $\boldsymbol{D}_{i-1}^{\zeta}$. $\boldsymbol{\zeta}_j$ is row j of the matrix $\boldsymbol{\zeta}$ and $D_{i-1,j}^{\zeta}$ is the jth element in $\boldsymbol{D}_{i-1}^{\zeta}$. Therefore for $\mathbb{P} \in \mathcal{P}$, we have

$$\begin{aligned} &\inf_{\mathbb{P}\in\mathcal{P}} \Pr_{[\mathbb{P}]}\{\boldsymbol{U}_i \text{ is ineasible for (46) }\} \\ &\leq \Pr\left\{\boldsymbol{\zeta}_j\boldsymbol{D}_{i-1}^{\zeta} > \Omega_1\sqrt{(D_{i-1,j}^{\zeta})^2}\right\} \leq \exp\left\{-\Omega_1^2/2\right\}, \end{aligned} \quad (50)$$

where $j = 1, \ldots, n_D$. □

Similar to Theorem 2, for each DRMPC controller of vehicle i, the control chance constraints (25) has the following robust counterpart approximations framework under the Ball convex set:

$$\begin{cases} \boldsymbol{G}_a(\bar{\boldsymbol{K}}\boldsymbol{M}_{\boldsymbol{\varphi}}\bar{\boldsymbol{x}}_i(k|k) + \bar{\boldsymbol{K}}\boldsymbol{M}_B\boldsymbol{U}_i + \boldsymbol{U}_i) \leq \boldsymbol{f}_a \\ -\boldsymbol{M}_C\boldsymbol{D}_{i-1}^0 - \boldsymbol{M}_C\boldsymbol{\Omega}_2\sqrt{(\boldsymbol{D}_{i-1}^{\zeta})^2}; \\ \boldsymbol{\Omega}_2 \geq \sqrt{2\ln(1/\epsilon_a)} \otimes \mathbf{1}_{K\times K}. \end{cases} \quad (51)$$

C. The Algorithm of DRMPC-M2

Using the counterpart form of the objective and the safe convex approximations of the chance constraints, (45) and (51), **DRMPC-M1** can be reformulated as the following conic optimization model **DRMPC-M2**:

$$\begin{aligned} \min_{\boldsymbol{U}_i}\ & U_i^{\mathrm{T}}\Theta U_i + \mathbf{h}^{\mathrm{T}}U_i + s \\ \text{s.t.}\ & (45), (51), i \in \mathcal{I}_N \\ & \bar{\boldsymbol{x}}_i(k+K|k) \in \Theta_K. \end{aligned} \quad (52)$$

The model (52) can be used to determine if this quadratic programming model is a computationally tractable convex optimization model. The model is solved by some off-the-shelf convex optimization techniques, such as sequential quadratic programming method [42]. Based on the model (52), the implementation of the DRMPC algorithm is summarized in Algorithm 1. Figure 1 also shows the implementation of the control method, which follows a distributed computing paradigm. During the platooning process, every following vehicle deploys a controller, which is responsible for solving the closed-loop DRMPC platoon model and implementing the solved control signal to each following vehicle.

Algorithm 1: The algorithm design for **DRMPC-M2**

Input: The prespecified small tolerances ϵ_a, ϵ_x, the initial state $\boldsymbol{x}_i(k|k)$ and the prediction time horizon K for vehicle i;
Output: the control signal $a_i^*(k|k)$ and $\bar{\boldsymbol{x}}_i(k+1|k)$
Initialization: $\bar{\boldsymbol{x}}_i(k|k) = \boldsymbol{x}_i(k|k)$;
Solve an LQR problem to obtain $\boldsymbol{K}$;
Solve **DRMPC-M2** to obtain optimal $\boldsymbol{U}_i^*$;
Implement the first element $u_i^*(k|k)$ from $\boldsymbol{U}_i^*$;
Compute the control signal $a_i^*(k|k) = \boldsymbol{K}\boldsymbol{x}_i(k|k) + u_i^*(k|k)$;
Compute $\bar{\boldsymbol{x}}_i(k+1|k)$ according to (18).

D. Recursive Feasibility and Convergence Analysis

This section discusses the recursive feasibility and convergence of Algorithm 1 and provides proofs in the Appendix. The recursive feasibility of Model **DRMPC-M2** is introduced through Theorem 3. Then, we can analyse its convergence according to the feasibility conclusion Theorem 4.

Theorem 3: There exists a feasible solution at time $k = 0$ for the model **DRMPC-M2**, such that this model is recursively feasible under Algorithm 1.

Proof: Please refer to Appendix B. □

According to [17], the chance constraints (24) and (25) are valid under the most recent time step τ when the control algorithm is feasible under feedback. Specifically, when the distribution information of the disturbance vector is $\mathcal{P}$ and condition $\bar{\boldsymbol{x}}_i(k|k) = \boldsymbol{x}_i(k|k)$ is satisfied, then the chance constraints is implicitly enforced when $\tau = k$.

In this paper, we propose a closed-loop DRMPC framework for the vehicle platoon problem with stochastic disturbances, where the basic idea of implementing closed-loop control is to map the system state onto the control sequence. Thus we propose the linear feedback strategy $a_i(k|k) = \boldsymbol{K}\boldsymbol{x}_i(k|k) + u_i(k|k)$, which parameterises the control of the vehicle by linearly combining the system's state. with the actual state of the vehicle at the current moment being fed back to the measurement system for optimisation for the next prediction. Then, the current state of the vehicle (position, velocity) is feedback to the measurement system to update the state of the formation system which is required for the next prediction $\boldsymbol{x}_i(k+1|k)$. Next, we provide the convergence of the **DRMPC-M2** algorithm under the closed-loop system (18).

Theorem 4: The model **DRMPC-M2** in a closed-loop system (18) with a feedback control strategy (17) is convergent if this system (18) satisfies the following condition:

$$\lim_{\tilde{K}\to\infty} \frac{1}{\tilde{K}} \sum_{k=0}^{\tilde{K}} \mathbb{E}_{[\mathbb{P}]} \left\{ \boldsymbol{x}_i(k|k)^{\mathrm{T}} \boldsymbol{Q} \boldsymbol{x}_i(k|k) \right\} \leq \mathrm{tr} \left\{ \boldsymbol{C}^{\mathrm{T}} \boldsymbol{Q}_K \boldsymbol{C} \Sigma \right\}. \tag{53}$$

Proof: Please refer to Appendix C. □

From Theorem 3 and Theorem 4, the recursive feasibility and the convergence of Algorithm 1 depends on the construction of the ambiguity set Ball_Ω and terminal constraint set $\bar{\boldsymbol{x}}_i(k+K|k) \in \Theta_K$.

E. Stability Analysis

In this section, we present the stability analysis method of the proposed DRMPC model. Specifically, we use the discrete Fourier transform (DFT) to measure the oscillation amplitude of the vehicle spacing error and perform a string stability analysis of the platoon system [43]. As stochastic disturbances in V2V communication are introduced into the state equation, string stability analysis cannot be performed well in the time domain under dynamic random disturbances. Therefore, we use a frequency domain method to assess local string stability under stochastic disturbances. In this paper, we apply the DFT function to the spacing error of the following vehicle to assess the stability of the platoon system in the frequency domain. In this paper, the spacing errors of two consecutive vehicles $i-1$ and i are calculated from the observed and expected distances d_i. The spacing error in the time domain has the following form

$$\triangle s_i(k) = \triangle p_i(k) - d_i. \tag{54}$$

Then, we apply the Fourier transform to the spacing error function $s_i(k)$ to obtain the transfer function of the spacing error in the frequency domain

$$\delta_i(\rho) = \sum_k \triangle s_i(k) e^{-j\rho k}, \forall \rho \in [0, \pi], j = \sqrt{-1}. \tag{55}$$

TABLE I
PARAMETERS SETTINGS

Parameter	Value	Parameter	Value
ϵ_x	0.05	ϵ_a	0.05
N	5	τ	0.01 s
r	0.01 s	K	20 s
L	2 m	d_i	10 m
$p_{\min}$	0 m	$p_{\max}$	+3 m
$v_{\min}$	−3 m/s	$v_{\max}$	+3 m/s
$a_{\min}$	$-1.5\,\mathrm{m/s^2}$	$a_{\max}$	$+1.5\,\mathrm{m/s^2}$

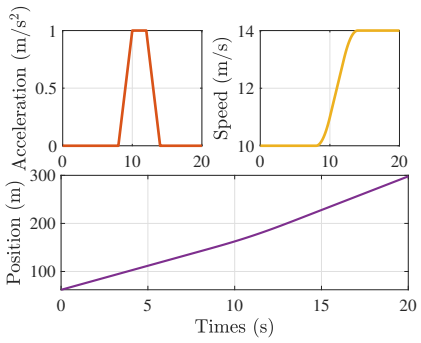


Fig. 3. The motion trajectory of the leading vehicle.

The string stability of the controller in the platoon vehicle system is concerned with the increment and decay of the signal (spacing error). We evaluate the string stability of the controlled vehicle in the frequency domain by the magnitude of the transfer function [44]. Based on equation (55), we propose the following metric function

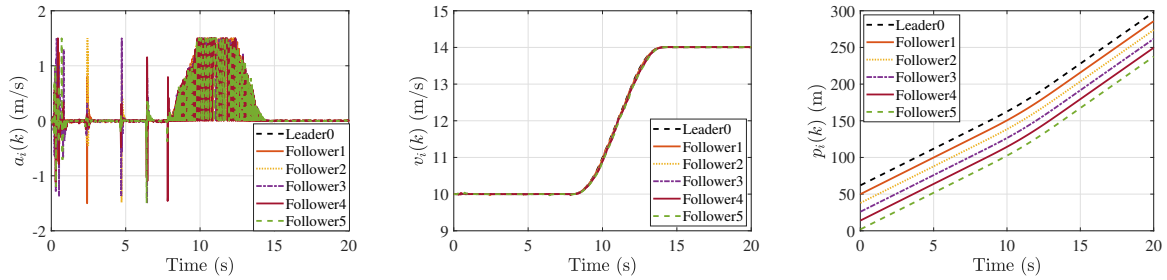
$$\Psi_i(\rho) = \left| \frac{\delta_i(\rho)}{\delta_{i-1}(\rho)} \right|, \forall \rho \in [0, \pi]. \tag{56}$$

where $\delta_i(\rho)$ and $\delta_{i-1}(\rho)$ are the Fourier transform of spacing error $\triangle s_i(k)$ between the controlled vehicle i and the front vehicle $i-1$. If $\Psi_i(\rho)$ is decreasing along the platoon, then the platooning system under the DRMPC controller is string stable in the frequency domain. The analysis process is presented in Appendix D.

IV. Simulation

Simulations are conducted in a specific platoon system with five following vehicles (indexed from 1 to 5) and one leading vehicle (indexed by 0) to verify the effectiveness of the control method under model **DRMPC-M2**. In addition, all simulations are implemented in Matbal/Simulink-based solver version R2021a running on a 2.8-GHz 64-bit Core i7-8400U CPU machine under Windows 10 Professional. In addition, the entire simulation takes 20 s and the predicted time $K = 0.05$ s.

Figure 3 shows the trajectory of the leading vehicle in the platoon system. This leading vehicle is uniformly accelerated and decelerated between [8, 14] s, and then moves at a constant speed $v = 14\,\mathrm{m/s}$ during the other phases. The initial state information of the lead vehicle is $p_0(0) = 60\,\mathrm{m}$ and $v_0(0) = 10\,\mathrm{m/s}$. The initial longitudinal position of all subsequent vehicles is within the range [0, 100] m with a minimum



(a) Acceleration of the vehicles in platoon (b) Speed of the vehicles in platoon (c) Position of the vehicles in platoon

(d) Speed error of the vehicles in platoon (e) Spacing error of the vehicles in platoon

Fig. 4. The acceleration, speed, position, speed error and spacing error of the vehicles with the proposed DRMPC platoon controller.

distance of 12 m between vehicles. In our setting, the feedback control gain $\boldsymbol{K}$ is obtained offline, i.e., $\boldsymbol{K} = [0.0096, 0.1103]^{\mathrm{T}}$. The other parameter setting is summarized in Table I.

A. Performance Verification

Figures 4(a)-4(e) show the results of the proposed DRMPC model. Figure 4(a)-4(c) shows the acceleration, speed and longitudinal position of the vehicles in the platoon. As can be seen, with the designed controller, all control inputs (acceleration/deceleration) are strictly limited between the interval [-1.5, 1.5] $\mathrm{m/s^2}$ and each following vehicle can track the speed of the previous vehicle. Even in the presence of external disturbances, all vehicles form a stable platoon without any collisions. Figures 4(d) and 4(e) show the spacing and speed errors of two neighbouring vehicles in the formation. As can be seen in Figure 4(d), all the following vehicles can track the speed of the leading vehicle. Specifically, all vehicles can track the desired speed $v = 14\,\mathrm{m/s}$ stably after about 14 s. In addition, as can be seen in Figure 4(e), the proposed DRMPC controller guarantees the safety of the platoon and increases the spacing between two adjacent vehicles to withstand random disturbances to enhance the robustness of the controller when the acceleration of the leading vehicle suddenly changes.

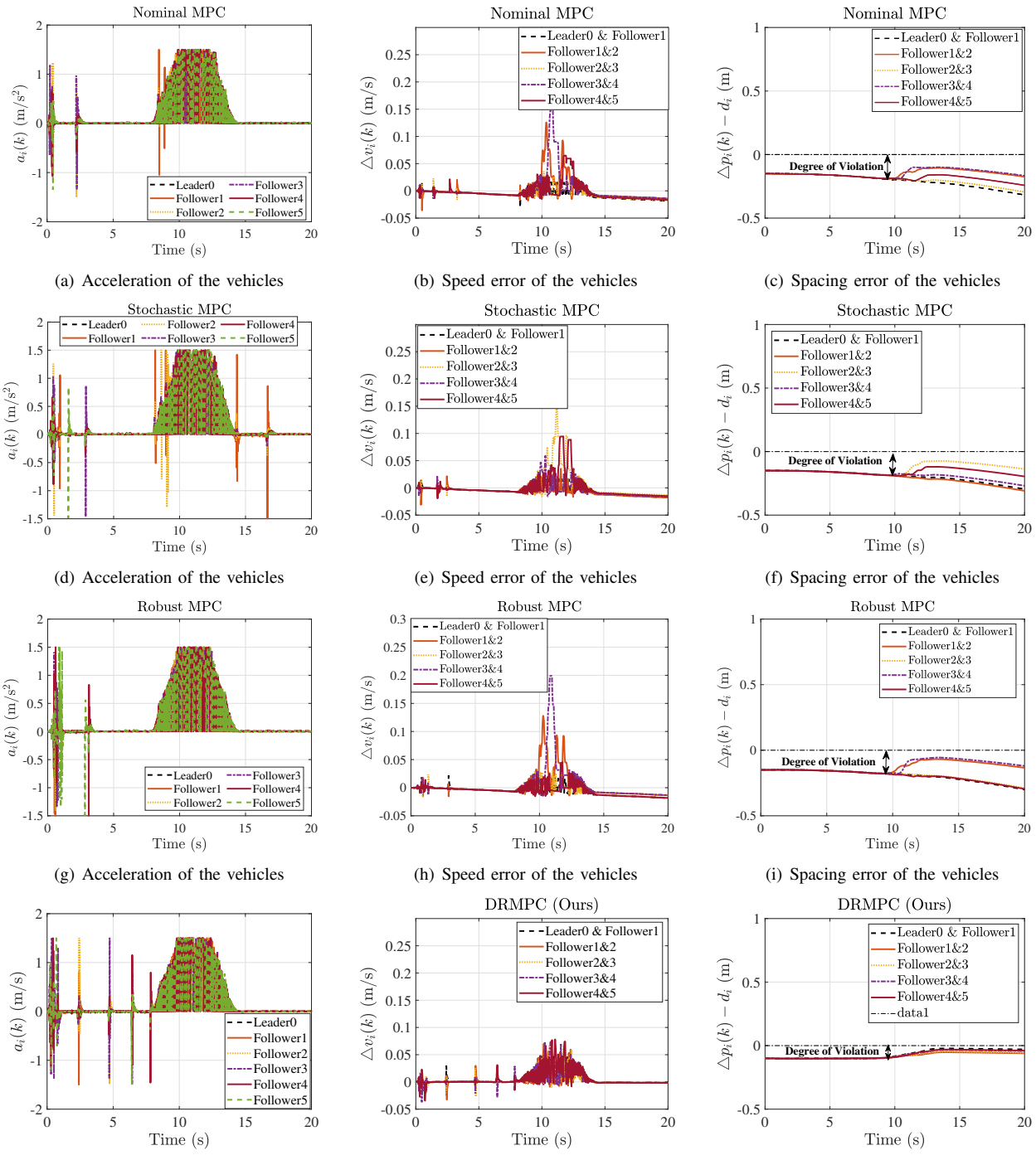
B. Performance Comparison

In this Section, we compare the proposed controller with three traditional controllers under three different types of disturbances. The four types of vehicle controllers are:

- **Nominal MPC**: Nominal MPC controllers are not resistant to stochastic interference, they only make decisions for the nominal terms of the system.
- **Stochastic MPC**: In the SMPC controller, we adopt the state-of-the-art Chebyshev-Cantilli method to deal with the probabilistic constraint [45].
- **Robust MPC**: In RMPC controller, the stochastic variable ζ_k^l belongs to the bounded set. This controller obtains the vehicle acceleration based on the worst-case scenario principle to resist bounded disturbances.
- **Proposed DRMPC(Ours)**: We design a DRMPC controller for the platooning vehicle. This controller under the Ball ambiguity set can resist stochastic disturbances under different probability distributions.

In addition, we compare the control performance of the four methods under two different probability distributions. The first probability distribution is that the stochastic variable $\boldsymbol{\zeta}$ obeys a Gaussian distribution, i.e., $\boldsymbol{\zeta} \sim (0.0001, 0.003)$. The second is that the stochastic variable $\boldsymbol{\zeta}$ obeys a Uniform distribution, i.e., $\boldsymbol{\zeta} \sim [-0.15, 0.15]$. For the Ball ambiguity set we choose $\epsilon_x = \epsilon_a = 0.05$ and $\Omega_1 = \Omega_2 = 2.44$. Other parameters are set as shown in Table I. We evaluate the performance of the four control methods in terms of safety (spacing error and speed error). The acceleration, speed error and spacing error of the 5 following vehicles are shown in Figure 5, and the proposed DRMPC controller performs well in both safety and robustness. As can be seen from subfigures 5(a)-5(c), the NMPC method is the weakest against stochastic perturbations. The method is unable to guarantee the consistency of the speeds of all the vehicles and its probability of violation constraints continues to increase. It can be seen from subfigure 5(c) that the spacing between two neighbouring vehicles under this control method is getting smaller and smaller, which indicates that there is a risk of collision in the platooning system under the NMPC method.

All three methods SMPC, RMPC and DRMPC can cope



(a) Acceleration of the vehicles (b) Speed error of the vehicles (c) Spacing error of the vehicles

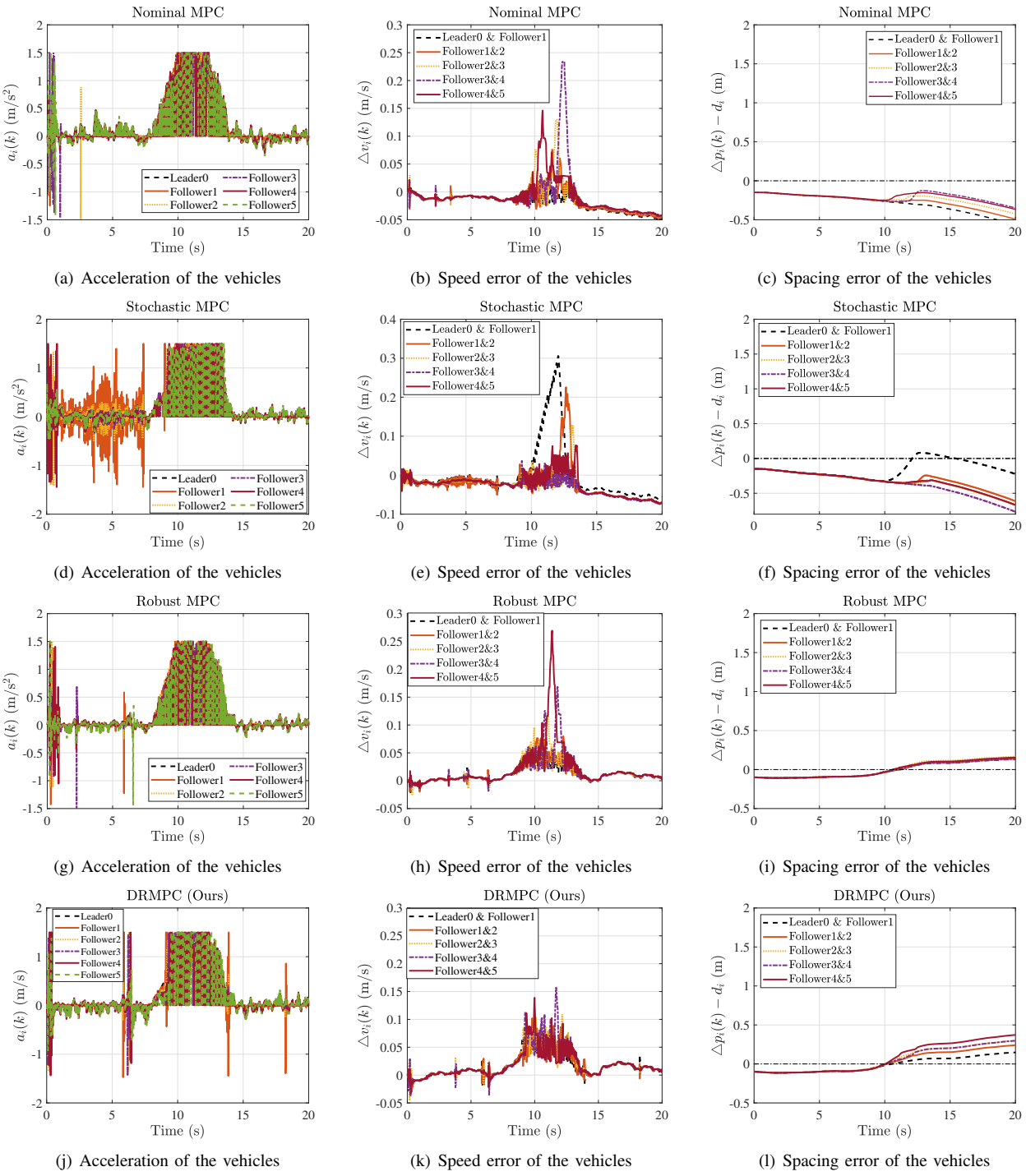
(d) Acceleration of the vehicles (e) Speed error of the vehicles (f) Spacing error of the vehicles

(g) Acceleration of the vehicles (h) Speed error of the vehicles (i) Spacing error of the vehicles

(j) Acceleration of the vehicles (k) Speed error of the vehicles (l) Spacing error of the vehicles

Fig. 5. The acceleration, speed error and spacing error of the vehicles with four controllers under Gaussian distribution.

with stochastic perturbations in the platoon system. When the stochastic perturbations belong to a uniform distribution, the safety of the platoon system under the RMPC and DRMPC methods is higher than that of the SMPC method, because the spacing error of the four following vehicles under the SMPC method is continuously decreasing, and the degree of violating probabilistic constraints is increasing, which indicates that the probability that the security of the platoon may be threatened at some point in the future is increased [subfigure 5(f)]. In addition, it can be seen from subfigures 5(i) and 5(l) that the correction ability (robustness) of DRMPC is higher than that of other control methods under small perturbations. In addition, we find it interesting that the vehicle platoon violates the lower bound constraint $\triangle p_i(k) - d_i = 0\ m$ when the vehicle spacing error is just starting to travel. This is reasonable due to the probabilistic chance constraints that we have introduced into the design of the model, which allows the model to allow for the probability of a violation. This also demonstrates the superiority of the proposed controller in that when the lead vehicle starts accelerating/decelerating, the proposed DRMPC



(a) Acceleration of the vehicles (b) Speed error of the vehicles (c) Spacing error of the vehicles

(d) Acceleration of the vehicles (e) Speed error of the vehicles (f) Spacing error of the vehicles

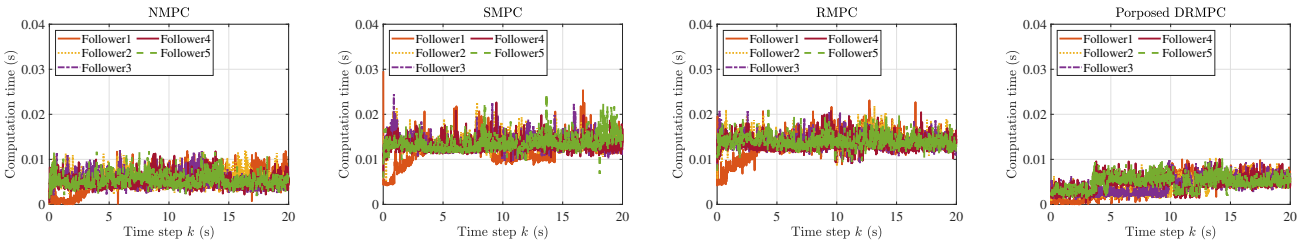
(g) Acceleration of the vehicles (h) Speed error of the vehicles (i) Spacing error of the vehicles

(j) Acceleration of the vehicles (k) Speed error of the vehicles (l) Spacing error of the vehicles

Fig. 6. The acceleration, speed error and spacing error of the vehicles with four controllers under Uniform distribution

can reduce the degree of violation probability to safeguard the platoon's safety and robustness.

We provide statistics such as the average spacing error in Table II to show the difference between the four control methods. The bold numbers in Table II indicate that the DRMPC method has better control performance than NMPC, SMPC and RMPC. According to Table II, although all three controllers are resistant to random disturbances to a certain extent, the proposed DRMPC has better control performance than other control methods. For the following vehicles 1-5, the DRMPC method has the largest average spacing error in the vehicle platoon, which implies the enhanced safety and robustness of the vehicles in the platoon. For example, the average spacing error for following vehicle 5 is -0.071 m, which improves the control performance by 55.06% compared to SMPC and 63.21% compared to RMPC. By comparison, the DRMPC method is more resistant to interference and ensures that the platooning vehicles travel safely despite the constant interference to the system.



(a) Execution time of NMPC (b) Execution time of SMPC (c) Execution time of RMPC (d) Execution time of DRMPC

Fig. 7. The computational efficiency of four control methods.

TABLE II
THE AVERAGE SPACING ERROR UNDER GAUSSIAN DISTRIBUTION

Method \ Vehicle	0&1	1&2	2&3	3&4	4&5
NMPC	-0.206	-0.149	-0.197	-0.145	-0.178
SMPC	-0.196	-0.202	-0.135	-0.185	-0.158
RMPC	-0.194	-0.128	-0.189	-0.122	-0.193
DRMPC	**-0.066**	**-0.080**	**-0.078**	**-0.072**	**-0.071**

TABLE III
THE AVERAGE SPACING ERROR UNDER UNIFORM DISTRIBUTION

Method \ Vehicle	0&1	1&2	2&3	3&4	4&5
NMPC	-0.289	-0.264	-0.233	-0.211	-0.210
SMPC	-0.152	-0.309	-0.330	-0.362	-0.328
RMPC	-0.007	-0.005	0.006	-0.005	0.001
DRMPC	**-0.007**	**0.028**	**0.052**	**0.049**	**0.080**

On the other hand, we compare the control performance of the four methods in the presence of a large range of uncertain disturbances ($\boldsymbol{\zeta}$ belonging to a uniform distribution). Figure 6 shows the acceleration, velocity error and spacing error under the four control methods. Similarly, the safety of the formation vehicles is threatened under the NMPC control method, and the spacing between two neighbouring vehicles gradually decreases. From subfigures 6(f), 6(i) and 6(l), it can be seen that the proposed DRMPC method increases the vehicle spacing under stochastic perturbation compared to the RMPC and SMPC methods, which indicates that the safety and robustness of the formation are improved. Likewise, to show the subtle differences between the different methods, Table III provides the average spacing errors of the 5 following vehicles. Similarly, the average spacing of the platoon vehicle under the DRMPC method is larger than the other three methods. We also found that although the RMPC method also has the ability to withstand uncertain disturbances, DRMPC has the lowest vehicle collision risk under both Gaussian distribution and Uniform distribution, which shows that the DRMPC method is better than the other disturbance-resistant methods in terms of robustness and safety.

V. CONCLUSION AND FUTURE WORK

We have proposed a DRMPC platoon control method for a string of CAVs with V2V communication uncertainty. First, we established uncertain state equations for each pair of neighbouring vehicles in the platoon, then we developed an MPC-based distributionally robust optimization model with an unknown probability distribution for the vehicle platoon by imposing chance constraints on state and control and using the expectation value of the cost function as the objective value. To solve the proposed DRMPC model, we applied the robust approximation theory to reconstruct the robust counterpart framework of DRMPC and then designed the DRMPC algorithm and proved the recursive feasibility and convergence. To verify the effectiveness of the model, we compared the proposed DRMPC method with NMPC, RMPC and SMPC methods, and conducted comparative experiments to explore the optimal trade-offs. Through the results of simulation experiments and performance comparisons, we verify the advantages of the proposed control method in achieving the platoon objectives, resisting stochastic perturbations, and platoon stability. In future work, we will explore the robust control method under hybrid vehicle formation and analyse the influence of driver behaviour on the formation effect.

APPENDIX A

In our paper, the objective function has the following form:

$$\begin{aligned} J(\boldsymbol{X}_i,\boldsymbol{U}_i) =&\mathbb{E}_{[\mathbb{P}]}\Bigg[\sum_{h=0}^{K-1} f(\boldsymbol{x}_i(k+h|k),a_i(k+h|k)) \\ &+\Phi(\boldsymbol{x}_i(k+K|k))\Bigg] \\ =&\mathbb{E}_{[\mathbb{P}]}\left\{\boldsymbol{X}_i^{\mathrm{T}}\boldsymbol{Q}_x\boldsymbol{X}_i+\boldsymbol{a}_i^{\mathrm{T}}\boldsymbol{R}_u\boldsymbol{a}_i\right\} \\ =&\mathbb{E}_{[\mathbb{P}]}\left\{\boldsymbol{X}_i^{\mathrm{T}}\boldsymbol{Q}_x\boldsymbol{X}_i\right\}+\mathbb{E}_{[\mathbb{P}]}\left\{\boldsymbol{a}_i^{\mathrm{T}}\boldsymbol{R}_u\boldsymbol{a}_i\right\}. \end{aligned} \tag{57}$$

Based on the equation (22), we have

$$\begin{aligned} &\mathbb{E}_{[\mathbb{P}]}\left\{\boldsymbol{X}_i^{\mathrm{T}}\boldsymbol{Q}_x\boldsymbol{X}_i\right\} \\ =&\mathbb{E}_{[\mathbb{P}]}\Big\{\Big(\boldsymbol{M}_{\boldsymbol{\varphi}}\boldsymbol{x}_i(k|k)+\boldsymbol{M}_B\boldsymbol{U}_i+\boldsymbol{M}_C(\boldsymbol{D}_{i-1}^0+\boldsymbol{\zeta}\boldsymbol{D}_{i-1}^{\zeta})\Big)^{\mathrm{T}}\boldsymbol{Q}_x \\ &\Big(\boldsymbol{M}_{\boldsymbol{\varphi}}\boldsymbol{x}_i(k|k)+\boldsymbol{M}_B\boldsymbol{U}_i+\boldsymbol{M}_C(\boldsymbol{D}_{i-1}^0+\boldsymbol{\zeta}\boldsymbol{D}_{i-1}^{\zeta})\Big)\Big\} \\ =&\left(\boldsymbol{M}_{\boldsymbol{\varphi}}\mathbb{E}_{[\mathbb{P}]}\left\{\boldsymbol{x}_i(k|k)\right\}\right)^{\mathrm{T}}\boldsymbol{Q}_x\left(\boldsymbol{M}_{\boldsymbol{\varphi}}\mathbb{E}_{[\mathbb{P}]}\left\{\boldsymbol{x}_i(k|k)\right\}\right) \\ &+2\left(\boldsymbol{M}_{\boldsymbol{\varphi}}\mathbb{E}_{[\mathbb{P}]}\left\{\boldsymbol{x}_i(k|k)\right\}\right)^{\mathrm{T}}\boldsymbol{Q}_x\boldsymbol{M}_C\mathbb{E}_{[\mathbb{P}]}\left\{\boldsymbol{D}_{i-1}^0+\boldsymbol{\zeta}\boldsymbol{D}_{i-1}^{\zeta}\right\} \\ &+2\left(\boldsymbol{M}_{\boldsymbol{\varphi}}\mathbb{E}_{[\mathbb{P}]}\left\{\boldsymbol{x}_i(k|k)\right\}\right)^{\mathrm{T}}\boldsymbol{Q}_x\boldsymbol{M}_B\boldsymbol{U}_i+\boldsymbol{U}_i^{\mathrm{T}}\left(\boldsymbol{M}_B^{\mathrm{T}}\boldsymbol{Q}_x\boldsymbol{M}_B\right)\boldsymbol{U}_i \\ &+2\left(\boldsymbol{M}_C\mathbb{E}_{[\mathbb{P}]}\left\{\boldsymbol{D}_{i-1}^0+\boldsymbol{\zeta}\boldsymbol{D}_{i-1}^{\zeta}\right\}\right)^{\mathrm{T}}\boldsymbol{Q}_x\boldsymbol{M}_B\boldsymbol{U}_i \\ &+\mathbb{E}_{[\mathbb{P}]}\left\{\left(\boldsymbol{D}_{i-1}^0+\boldsymbol{\zeta}\boldsymbol{D}_{i-1}^{\zeta}\right)^{\mathrm{T}}\boldsymbol{M}_C^{\mathrm{T}}\boldsymbol{Q}_x\boldsymbol{M}_C\left(\boldsymbol{D}_{i-1}^0+\boldsymbol{\zeta}\boldsymbol{D}_{i-1}^{\zeta}\right)\right\} \end{aligned}$$

$$
\begin{aligned}
=&\bar{\boldsymbol{x}}_i^{\mathrm{T}}(k|k)\boldsymbol{M}_{\boldsymbol{\varphi}}^{\mathrm{T}}\boldsymbol{Q}_x\boldsymbol{M}_{\boldsymbol{\varphi}}\bar{\boldsymbol{x}}_i(k|k)\\
&+2\bar{\boldsymbol{x}}_i^{\mathrm{T}}(k|k)\boldsymbol{M}_{\boldsymbol{\varphi}}^{\mathrm{T}}\boldsymbol{Q}_x\boldsymbol{M}_C\boldsymbol{D}_{i-1}^0\\
&+2\bar{\boldsymbol{x}}_i^{\mathrm{T}}(k|k)\boldsymbol{M}_{\boldsymbol{\varphi}}^{\mathrm{T}}\boldsymbol{Q}_x\boldsymbol{M}_B\boldsymbol{U}_i+\boldsymbol{U}_i^{\mathrm{T}}\boldsymbol{M}_B^{\mathrm{T}}\boldsymbol{Q}_x\boldsymbol{M}_B\boldsymbol{U}_i\\
&+2\left(\boldsymbol{D}_{i-1}^0\right)^{\mathrm{T}}\boldsymbol{M}_C^{\mathrm{T}}\boldsymbol{Q}_x\boldsymbol{M}_B\boldsymbol{U}_i\\
&+(\boldsymbol{D}_{i-1}^0)^{\mathrm{T}}\boldsymbol{M}_C^{\mathrm{T}}\boldsymbol{Q}_x\boldsymbol{M}_C\boldsymbol{D}_{i-1}^0+(\boldsymbol{D}_{i-1}^{\boldsymbol{\zeta}})^{\mathrm{T}}\mathrm{tr}\left\{\boldsymbol{M}_C^{\mathrm{T}}\boldsymbol{Q}_x\boldsymbol{M}_C\boldsymbol{\Sigma}\right\}\boldsymbol{D}_{i-1}^{\boldsymbol{\zeta}}.
\end{aligned}
$$

Since $\mathbb{E}_{[\mathbb{P}]}\{\boldsymbol{\zeta}\}=\mathbf{0}$ and $\mathbb{E}_{[\mathbb{P}]}\{\boldsymbol{\zeta}\boldsymbol{\zeta}^{\mathrm{T}}\}=\boldsymbol{\Sigma}$, then the last term $\mathbb{E}_{[\mathbb{P}]}\left\{\boldsymbol{X}_i^{\mathrm{T}}\boldsymbol{Q}_x\boldsymbol{X}_i\right\}$ of can be simplified as

$$
\begin{aligned}
&\mathbb{E}_{[\mathbb{P}]}\left\{\left(\boldsymbol{D}_{i-1}^0+\boldsymbol{\zeta}\boldsymbol{D}_{i-1}^{\boldsymbol{\zeta}}\right)^{\mathrm{T}}\boldsymbol{M}_C^{\mathrm{T}}\boldsymbol{Q}_x\boldsymbol{M}_C\left(\boldsymbol{D}_{i-1}^0+\boldsymbol{\zeta}\boldsymbol{D}_{i-1}^{\boldsymbol{\zeta}}\right)\right\}\\
=&(\boldsymbol{D}_{i-1}^0)^{\mathrm{T}}\boldsymbol{M}_C^{\mathrm{T}}\boldsymbol{Q}_x\boldsymbol{M}_C\boldsymbol{D}_{i-1}^0+(\boldsymbol{D}_{i-1}^0)^{\mathrm{T}}\boldsymbol{M}_C^{\mathrm{T}}\boldsymbol{Q}_x\boldsymbol{M}_C\mathbb{E}_{[\mathbb{P}]}\{\boldsymbol{\zeta}\}\boldsymbol{D}_{i-1}^{\boldsymbol{\zeta}}\\
&+(\mathbb{E}_{[\mathbb{P}]}\{\boldsymbol{\zeta}\}\boldsymbol{D}_{i-1}^{\boldsymbol{\zeta}})^{\mathrm{T}}\boldsymbol{M}_C^{\mathrm{T}}\boldsymbol{Q}_x\boldsymbol{M}_C\boldsymbol{D}_{i-1}^0\\
&+\mathbb{E}_{[\mathbb{P}]}\left\{(\boldsymbol{\zeta}\boldsymbol{D}_{i-1}^{\boldsymbol{\zeta}})^{\mathrm{T}}\boldsymbol{M}_C^{\mathrm{T}}\boldsymbol{Q}_x\boldsymbol{M}_C(\boldsymbol{\zeta}\boldsymbol{D}_{i-1}^{\boldsymbol{\zeta}})\right\}\\
=&(\boldsymbol{D}_{i-1}^0)^{\mathrm{T}}\boldsymbol{M}_C^{\mathrm{T}}\boldsymbol{Q}_x\boldsymbol{M}_C\boldsymbol{D}_{i-1}^0+(\boldsymbol{D}_{i-1}^{\boldsymbol{\zeta}})^{\mathrm{T}}\mathrm{tr}\left\{\boldsymbol{M}_C^{\mathrm{T}}\boldsymbol{Q}_x\boldsymbol{M}_C\boldsymbol{\Sigma}\right\}\boldsymbol{D}_{i-1}^{\boldsymbol{\zeta}}.
\end{aligned}
$$

Besides, the control item of objective has the following form

$$
\begin{aligned}
&\mathbb{E}_{[\mathbb{P}]}\left\{\boldsymbol{a}_i^{\mathrm{T}}\boldsymbol{R}_u\boldsymbol{a}_i\right\}\\
=&\mathbb{E}_{[\mathbb{P}]}\left\{(\bar{\boldsymbol{K}}\boldsymbol{X}_i+\boldsymbol{U}_i)^{\mathrm{T}}\boldsymbol{R}_u(\bar{\boldsymbol{K}}\boldsymbol{X}_i+\boldsymbol{U}_i)\right\}\\
=&\mathbb{E}_{[\mathbb{P}]}\left\{\left[\bar{\boldsymbol{K}}\boldsymbol{M}_{\boldsymbol{\varphi}}\boldsymbol{x}_i(k|k)+\bar{\boldsymbol{K}}\boldsymbol{M}_B\boldsymbol{U}_i+\boldsymbol{U}_i+\bar{\boldsymbol{K}}\boldsymbol{M}_C\boldsymbol{D}_{i-1}^0\right]^{\mathrm{T}}\right.\\
&\left.\boldsymbol{R}_u\left[\bar{\boldsymbol{K}}\boldsymbol{M}_{\boldsymbol{\varphi}}\boldsymbol{x}_i(k|k)+\bar{\boldsymbol{K}}\boldsymbol{M}_B\boldsymbol{U}_i+\boldsymbol{U}_i+\bar{\boldsymbol{K}}\boldsymbol{M}_C\boldsymbol{D}_{i-1}^0\right]\right\}\\
&+\mathbb{E}_{[\mathbb{P}]}\left\{(\boldsymbol{\zeta}\boldsymbol{D}_{i-1}^{\boldsymbol{\zeta}})^{\mathrm{T}}(\bar{\boldsymbol{K}}\boldsymbol{M}_C)^{\mathrm{T}}\boldsymbol{R}_u(\bar{\boldsymbol{K}}\boldsymbol{M}_C)(\boldsymbol{\zeta}\boldsymbol{D}_{i-1}^{\boldsymbol{\zeta}})\right\}.
\end{aligned}
\tag{58}
$$

Similarly, the first term of (58) can be simplified as

$$
\begin{aligned}
&\mathbb{E}_{[\mathbb{P}]}\left\{\left[\bar{\boldsymbol{K}}\boldsymbol{M}_{\boldsymbol{\varphi}}\boldsymbol{x}_i(k|k)+\bar{\boldsymbol{K}}\boldsymbol{M}_B\boldsymbol{U}_i+\boldsymbol{U}_i+\bar{\boldsymbol{K}}\boldsymbol{M}_C\boldsymbol{D}_{i-1}^0\right]^{\mathrm{T}}\right.\\
&\left.\boldsymbol{R}_u\left[\bar{\boldsymbol{K}}\boldsymbol{M}_{\boldsymbol{\varphi}}\boldsymbol{x}_i(k|k)+\bar{\boldsymbol{K}}\boldsymbol{M}_B\boldsymbol{U}_i+\boldsymbol{U}_i+\bar{\boldsymbol{K}}\boldsymbol{M}_C\boldsymbol{D}_{i-1}^0\right]\right\}\\
=&\mathbb{E}_{[\mathbb{P}]}\left\{\left[\bar{\boldsymbol{K}}\boldsymbol{M}_{\boldsymbol{\varphi}}\boldsymbol{x}_i(k|k)+\bar{\boldsymbol{K}}\boldsymbol{M}_B\boldsymbol{U}_i+\boldsymbol{U}_i\right]^{\mathrm{T}}\boldsymbol{R}_u\right.\\
&\left.\left[\bar{\boldsymbol{K}}\boldsymbol{M}_{\boldsymbol{\varphi}}\boldsymbol{x}_i(k|k)+\bar{\boldsymbol{K}}\boldsymbol{M}_B\boldsymbol{U}_i+\boldsymbol{U}_i\right]\right\}\\
&+\mathbb{E}_{[\mathbb{P}]}\left\{\left[\bar{\boldsymbol{K}}\boldsymbol{M}_{\boldsymbol{\varphi}}\boldsymbol{x}_i(k|k)+\bar{\boldsymbol{K}}\boldsymbol{M}_B\boldsymbol{U}_i+\boldsymbol{U}_i\right]^{\mathrm{T}}\boldsymbol{R}_u\bar{\boldsymbol{K}}\boldsymbol{M}_C\boldsymbol{D}_{i-1}^0\right\}\\
&+\mathbb{E}_{[\mathbb{P}]}\left\{(\bar{\boldsymbol{K}}\boldsymbol{M}_C\boldsymbol{D}_{i-1}^0)^{\mathrm{T}}\boldsymbol{R}_u\left[\bar{\boldsymbol{K}}\boldsymbol{M}_{\boldsymbol{\varphi}}\boldsymbol{x}_i(k|k)+\bar{\boldsymbol{K}}\boldsymbol{M}_B\boldsymbol{U}_i+\boldsymbol{U}_i\right]\right\}\\
&+\mathbb{E}_{[\mathbb{P}]}\left\{(\bar{\boldsymbol{K}}\boldsymbol{M}_C\boldsymbol{D}_{i-1}^0)^{\mathrm{T}}\boldsymbol{R}_u(\bar{\boldsymbol{K}}\boldsymbol{M}_C\boldsymbol{D}_{i-1}^0)\right\}\\
=&\bar{\boldsymbol{x}}_i^{\mathrm{T}}(k|k)\boldsymbol{M}_{\boldsymbol{\varphi}}^{\mathrm{T}}\bar{\boldsymbol{K}}^{\mathrm{T}}\boldsymbol{R}_u\bar{\boldsymbol{K}}\boldsymbol{M}_{\boldsymbol{\varphi}}\bar{\boldsymbol{x}}_i(k|k)\\
&+2\bar{\boldsymbol{x}}_i^{\mathrm{T}}(k|k)\boldsymbol{M}_{\boldsymbol{\varphi}}^{\mathrm{T}}\bar{\boldsymbol{K}}^{\mathrm{T}}\boldsymbol{R}_u(\bar{\boldsymbol{K}}\boldsymbol{M}_B+\boldsymbol{I})\boldsymbol{U}_i\\
&+\boldsymbol{U}_i^{\mathrm{T}}(\bar{\boldsymbol{K}}\boldsymbol{M}_B+\boldsymbol{I})^{\mathrm{T}}\boldsymbol{R}_u(\bar{\boldsymbol{K}}\boldsymbol{M}_B+\boldsymbol{I})\boldsymbol{U}_i\\
&+2(\boldsymbol{D}_{i-1}^0)^{\mathrm{T}}(\bar{\boldsymbol{K}}\boldsymbol{M}_C)^{\mathrm{T}}\boldsymbol{R}_u\bar{\boldsymbol{K}}\boldsymbol{M}_{\boldsymbol{\varphi}}\bar{\boldsymbol{x}}_i(k|k)\\
&+2(\boldsymbol{D}_{i-1}^0)^{\mathrm{T}}(\bar{\boldsymbol{K}}\boldsymbol{M}_C)^{\mathrm{T}}\boldsymbol{R}_u(\bar{\boldsymbol{K}}\boldsymbol{M}_B+\boldsymbol{I})\boldsymbol{U}_i\\
&+(\boldsymbol{D}_{i-1}^0)^{\mathrm{T}}\boldsymbol{M}_C^{\mathrm{T}}\bar{\boldsymbol{K}}^{\mathrm{T}}\boldsymbol{R}_u\bar{\boldsymbol{K}}\boldsymbol{M}_C\boldsymbol{D}_{i-1}^0.
\end{aligned}
\tag{59}
$$

The second term of (58) can be simplified as

$$
\begin{aligned}
&\mathbb{E}_{[\mathbb{P}]}\left\{(\boldsymbol{\zeta}\boldsymbol{D}_{i-1}^{\boldsymbol{\zeta}})^{\mathrm{T}}(\bar{\boldsymbol{K}}\boldsymbol{M}_C)^{\mathrm{T}}\boldsymbol{R}_u(\bar{\boldsymbol{K}}\boldsymbol{M}_C)(\boldsymbol{\zeta}\boldsymbol{D}_{i-1}^{\boldsymbol{\zeta}})\right\}\\
=&(\boldsymbol{D}_{i-1}^{\boldsymbol{\zeta}})^{\mathrm{T}}\mathbb{E}_{[\mathbb{P}]}\left\{\boldsymbol{\zeta}^{\mathrm{T}}(\bar{\boldsymbol{K}}\boldsymbol{M}_C)^{\mathrm{T}}\boldsymbol{R}_u\bar{\boldsymbol{K}}\boldsymbol{M}_C\boldsymbol{\zeta}\right\}\boldsymbol{D}_{i-1}^{\boldsymbol{\zeta}}\\
=&(\boldsymbol{D}_{i-1}^{\boldsymbol{\zeta}})^{\mathrm{T}}\mathrm{tr}\left\{\boldsymbol{M}_C^{\mathrm{T}}\bar{\boldsymbol{K}}^{\mathrm{T}}\boldsymbol{R}_u\bar{\boldsymbol{K}}\boldsymbol{M}_C\boldsymbol{\Sigma}\right\}\boldsymbol{D}_{i-1}^{\boldsymbol{\zeta}}.
\end{aligned}
\tag{60}
$$

Then, we can obtain the robust counterpart form of expected objective:

$$
\begin{aligned}
&\Theta=\boldsymbol{M}_B^{\mathrm{T}}\boldsymbol{Q}_x\boldsymbol{M}_B+(\bar{\boldsymbol{K}}\boldsymbol{M}_B+\boldsymbol{I})^{\mathrm{T}}\boldsymbol{R}_u(\bar{\boldsymbol{K}}\boldsymbol{M}_B+\boldsymbol{I}).\\
&\mathbf{h}^{\mathrm{T}}=2\bar{\boldsymbol{x}}_i^{\mathrm{T}}(k|k)\boldsymbol{M}_{\boldsymbol{\varphi}}^{\mathrm{T}}\boldsymbol{Q}_x\boldsymbol{M}_B+2\left(\boldsymbol{D}_{i-1}^0\right)^{\mathrm{T}}\boldsymbol{M}_C^{\mathrm{T}}\boldsymbol{Q}_x\boldsymbol{M}_B\\
&\qquad+2\bar{\boldsymbol{x}}_i^{\mathrm{T}}(k|k)\boldsymbol{M}_{\boldsymbol{\varphi}}^{\mathrm{T}}\bar{\boldsymbol{K}}^{\mathrm{T}}\boldsymbol{R}_u(\bar{\boldsymbol{K}}\boldsymbol{M}_B+\boldsymbol{I})\\
&\qquad+2(\boldsymbol{D}_{i-1}^0)^{\mathrm{T}}\boldsymbol{M}_C^{\mathrm{T}}\bar{\boldsymbol{K}}^{\mathrm{T}}\boldsymbol{R}_u(\bar{\boldsymbol{K}}\boldsymbol{M}_B+\boldsymbol{I}.
\end{aligned}
\tag{61}
$$

$$
\begin{aligned}
s=&\bar{\boldsymbol{x}}_i^{\mathrm{T}}(k|k)\boldsymbol{M}_{\boldsymbol{\varphi}}^{\mathrm{T}}\boldsymbol{Q}_x\boldsymbol{M}_{\boldsymbol{\varphi}}\bar{\boldsymbol{x}}_i(k|k)+2\bar{\boldsymbol{x}}_i^{\mathrm{T}}(k)\boldsymbol{M}_{\boldsymbol{\varphi}}^{\mathrm{T}}\boldsymbol{Q}_x\boldsymbol{M}_C\boldsymbol{D}_{i-1}^0\\
&+(\boldsymbol{D}_{i-1}^0)^{\mathrm{T}}\boldsymbol{M}_C^{\mathrm{T}}\boldsymbol{Q}_x\boldsymbol{M}_C\boldsymbol{D}_{i-1}^0\\
&+(\boldsymbol{D}_{i-1}^{\boldsymbol{\zeta}})^{\mathrm{T}}\mathrm{tr}\left\{\boldsymbol{M}_C^{\mathrm{T}}\boldsymbol{Q}_x\boldsymbol{M}_C\boldsymbol{\Sigma}\right\}\boldsymbol{D}_{i-1}^{\boldsymbol{\zeta}}\\
&+\bar{\boldsymbol{x}}_i^{\mathrm{T}}(k|k)\boldsymbol{M}_{\boldsymbol{\varphi}}^{\mathrm{T}}\bar{\boldsymbol{K}}^{\mathrm{T}}\boldsymbol{R}_u\bar{\boldsymbol{K}}\boldsymbol{M}_{\boldsymbol{\varphi}}\bar{\boldsymbol{x}}_i(k|k)\\
&+2(\boldsymbol{D}_{i-1}^0)^{\mathrm{T}}\boldsymbol{M}_C^{\mathrm{T}}\bar{\boldsymbol{K}}^{\mathrm{T}}\boldsymbol{R}_u\bar{\boldsymbol{K}}\boldsymbol{M}_{\boldsymbol{\varphi}}\bar{\boldsymbol{x}}_i(k|k)\\
&+(\boldsymbol{D}_{i-1}^0)^{\mathrm{T}}\boldsymbol{M}_C^{\mathrm{T}}\bar{\boldsymbol{K}}^{\mathrm{T}}\boldsymbol{R}_u\bar{\boldsymbol{K}}\boldsymbol{M}_C\boldsymbol{D}_{i-1}^0\\
&+(\boldsymbol{D}_{i-1}^{\boldsymbol{\zeta}})^{\mathrm{T}}\mathrm{tr}\left\{\boldsymbol{M}_C^{\mathrm{T}}\bar{\boldsymbol{K}}^{\mathrm{T}}\boldsymbol{R}_u\bar{\boldsymbol{K}}\boldsymbol{M}_C\boldsymbol{\Sigma}\right\}\boldsymbol{D}_{i-1}^{\boldsymbol{\zeta}}.
\end{aligned}
\tag{62}
$$

APPENDIX B

To prove the recursive feasibility of the model, we need to show that there exists a feasible solution to the DRMPC model at time $k+1$ when the initial state satisfies $\bar{\boldsymbol{x}}_i(k+1|k+1)=\bar{\boldsymbol{x}}_i(k+1|k)$. First, we assume that the model is feasible at time k and the optimal solution is $\boldsymbol{U}_{i,k}^*=\{u_i^*(k+1|k),u_i^*(k+2|k),\ldots,u_i^*(k+K-1|k)\}$, then we need prove that $\boldsymbol{U}_{i,k+1}$ is feasible of the DRMPC model at time $k+1$, where $\boldsymbol{U}_{i,k+1}=\{u_i^*(k+1|k+1),u_i^*(k+2|k+1),\ldots,u_i^*(k+K-1|k+1),0\}$. If $\boldsymbol{U}_{i,k+1}$ is feasible for the DRMPC model, we can prove that this model is recursive feasibility.

Obviously, the solutions in $\boldsymbol{U}_{i,k+1}$ from $u_i^*(k+1|k+1)$ to $u_i^*(k+K-1|k+1)$ are feasible for the DRMPC. Second, we will verify that $\boldsymbol{U}_{i,k+1}$ is similarly feasible at time $k+K$ and the terminal constraint is satisfied. Based on Theorem 3 and 4, for each $k=0,1,\ldots,K-1$, the chance constraints of state and control can be rewritten as the (45) and (51). And at time $k+K$, the optimal solution $u_i^*(k+K|k+1)=0$ is also feasible for the proposed DRMPC model, and the terminal constraint condition satisfies $\bar{\boldsymbol{x}}_i(k+K+1)=\boldsymbol{\varphi}\bar{\boldsymbol{x}}_i(k+K|k)\in\Theta_K$. Therefore, we can deduce that the optimal control $\boldsymbol{U}_{i,k+1}$ is feasible for the DRMPC model **DRMPC-M2**.

APPENDIX C

Following the proof of Theorem 3, we assume that the optimal acceleration of DRMPC model at time $k+1$ is $\boldsymbol{a}_{i,k+1}^*=\{a_i^*(k+1|k+1),\ldots,a_i^*(k+K-1|k+1),a_i^*(k+K|k+1)\}$. Based on the terminal constraints (35), we construct a suboptimal acceleration under $\tilde{\boldsymbol{a}}_{i,k+1}$, which has the following form

$$
\tilde{\boldsymbol{a}}_{i,k+1}=\{a_i^*(k+1|k+1),\ldots,\boldsymbol{K}\boldsymbol{x}_i(k+K|k+1)\}.
\tag{63}
$$

At times $k+1$, the corresponding optimal objective is $J(\boldsymbol{X}_{i,k+1},\boldsymbol{a}_{i,k+1}^*)$ and we can obtain the relation

$$
J(\boldsymbol{X}_{i,k+1},\boldsymbol{a}_{i,k+1}^*)\leq J(\boldsymbol{X}_{i,k+1},\tilde{\boldsymbol{a}}_{i,k+1}).
\tag{64}
$$

Among them,

$$\begin{aligned}
&J(\boldsymbol{X}_{i,k+1},\tilde{\boldsymbol{a}}_{i,k+1})\\
=&J(\boldsymbol{X}_{i,k},\boldsymbol{a}^*_{i,k})-\mathbb{E}_{[\mathbb{P}]}\Big\{\boldsymbol{x}_i(k|k)^{\mathrm{T}}\boldsymbol{Q}\boldsymbol{x}_i(k|k)+a^*_i(k|k)^{\mathrm{T}}Ra^*_i(k|k)\Big\}\\
&+\mathbb{E}_{[\mathbb{P}]}\Big\{[\boldsymbol{K}\boldsymbol{x}_i(k+K|k)]^{\mathrm{T}}R[\boldsymbol{K}\boldsymbol{x}_i(k+K|k)]\Big\}\\
&+\mathbb{E}_{[\mathbb{P}]}\Big\{[\boldsymbol{K}\boldsymbol{x}_i(k+K+1|k)]^{\mathrm{T}}\boldsymbol{Q}_K[\boldsymbol{K}\boldsymbol{x}_i(k+K+1|k)]\Big\}\\
=&J(\boldsymbol{X}_{i,k},\boldsymbol{a}^*_{i,k})-\mathbb{E}_{[\mathbb{P}]}\Big\{\boldsymbol{x}_i(k|k)^{\mathrm{T}}\boldsymbol{Q}\boldsymbol{x}_i(k|k)+a^*_i(k|k)^{\mathrm{T}}Ra^*_i(k|k)\Big\}\\
&+\mathbb{E}_{[\mathbb{P}]}\Big\{\boldsymbol{x}_i(k+K|k)^{\mathrm{T}}\boldsymbol{Q}_Kx_i(k+K|k)\\
&+[\boldsymbol{K}\boldsymbol{x}_i(k+K|k)]^{\mathrm{T}}R[\boldsymbol{K}\boldsymbol{x}_i(k+K|k)]\Big\}\\
&-\mathbb{E}_{[\mathbb{P}]}\Big\{\boldsymbol{x}_i(k+K|k)^{\mathrm{T}}\boldsymbol{Q}_K\boldsymbol{x}_i(k+K|k)\\
&-[\boldsymbol{x}_i(k+K+1|k)]^{\mathrm{T}}\boldsymbol{Q}_K[\boldsymbol{x}_i(k+K+1|k)]\Big\},
\end{aligned}\tag{65}$$

where

$$\begin{aligned}
\boldsymbol{x}_i(k+K+1|k)=&\boldsymbol{A}\boldsymbol{x}_i(k+K|k)\\
&+\boldsymbol{B}a_i(k+K|k)+\boldsymbol{C}a_{i-1}(k+K|k).
\end{aligned}\tag{66}$$

When the platoon system is stable, the feedback control law is: $a_i(k)=\boldsymbol{K}\boldsymbol{x}_i(k)$, the terminal state (68) can be written as

$$\boldsymbol{x}_i(k+K+1|k)=(\boldsymbol{A}+\boldsymbol{B}\boldsymbol{K})\boldsymbol{x}_i(k+K|k)+\boldsymbol{C}a_{i-1}(k+K).\tag{67}$$

According to the algebraic Lyapunov equation (30), (65) can be conducted as

$$\begin{aligned}
&J(\boldsymbol{X}_{i,k+1},\tilde{\boldsymbol{a}}_{i,k+1})-J(\boldsymbol{X}_{i,k},\boldsymbol{a}^*_{i,k})\\
&=-\mathbb{E}_{[\mathbb{P}]}\Big\{\boldsymbol{x}_i(k|k)^{\mathrm{T}}\boldsymbol{Q}\boldsymbol{x}_i(k|k)+a^*_i(k|k)^{\mathrm{T}}Ra^*_i(k|k)\Big\}\\
&+\mathrm{tr}\{\boldsymbol{C}^{\mathrm{T}}\boldsymbol{Q}_K\boldsymbol{C}\Sigma\}.
\end{aligned}\tag{68}$$

Based on (64), we have

$$\begin{aligned}
&J(\boldsymbol{X}_{i,k+1},\boldsymbol{a}^*_{i,k+1})-J(\boldsymbol{X}_{i,k},\boldsymbol{a}^*_{i,k})\\
&\le J(\boldsymbol{X}_{i,k+1},\tilde{\boldsymbol{a}}_{i,k+1})-J(\boldsymbol{X}_{i,k},\boldsymbol{a}^*_{i,k}).
\end{aligned}\tag{69}$$

According to the reference [40], we take the limits on both sides of (69) as $k\to\infty$, we can obtain

$$\begin{aligned}
&\lim_{\tilde{K}\to\infty}\frac{1}{\tilde{K}}\sum_{k=0}^{\tilde{K}}\mathbb{E}_{[\mathbb{P}]}\left\{\boldsymbol{x}_i(k|k)^{\mathrm{T}}\boldsymbol{Q}\boldsymbol{x}_i(k|k)+a^*_i(k|k)^{\mathrm{T}}Ra^*_i(k|k)\right\}\\
&\le\mathrm{tr}\left\{\boldsymbol{C}^{\mathrm{T}}\boldsymbol{Q}_K\boldsymbol{C}\Sigma\right\}.
\end{aligned}$$

which implies $\lim_{\tilde{K}\to\infty}\frac{1}{\tilde{K}}\sum_{k=0}^{\tilde{K}}\mathbb{E}_{[\mathbb{P}]}\left\{\boldsymbol{x}_i(k|k)^{\mathrm{T}}\boldsymbol{Q}\boldsymbol{x}_i(k|k)\right\}\le\mathrm{tr}\left\{\boldsymbol{C}^{\mathrm{T}}\boldsymbol{Q}_K\boldsymbol{C}\Sigma\right\}$. The proof of Theorem 4 is completed.

APPENDIX D

Equation (56) is a metric function of vehicle spacing deviation in the frequency domain. Suppose $g_i(k)$ is the Laplace inverse transform of $\Psi_i(\rho)$, we have:

$$\begin{aligned}
\|\triangle s_i\|_\infty&=\max_k|\triangle s_i(k)|=\max_k\left|\int_0^k g_i(\tau)\triangle s_{i-1}(k-\tau)d\tau\right|\\
&\le\max_k\int_0^\infty|g_i(\tau)\|s_{i-1}(t-\tau)|\,d\tau\\
&\le\int_0^\infty|g_i(\tau)|d\tau\max_k|\triangle s_{i-1}(t-\tau)|=\|g_i\|_1\left\|\triangle s_{i-1}\right\|_\infty,
\end{aligned}$$

where $\|g_i\|_1=\int_0^\infty|g_i(\tau)|$. Meanwhile, if $\rho=j\omega$, we have by definition of Laplace transform that:

$$\Psi_i(jw)=\left|\int_0^\infty g_i(k)e^{-j\omega t}dt\right|=\int_0^\infty|g_i(t)|dt=\|g_i\|_1\tag{70}$$

According to literature [46], string stability requires that the spacing error along the vehicle string is decaying, i.e., $\|\triangle s_i\|_\infty<\|\triangle s_{i-1}\|_\infty<\ldots<\|\triangle s_1\|_\infty$, thus the sufficient condition for string stability of platoon system is:

$$\Psi_1(\rho)>\cdots>\Psi_{i-1}(\rho)>\Psi_i(\rho).\tag{71}$$

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Peiyu Zhang received her M.S. degree in mathematics from Hebei University, Baoding, China, in 2020. She is currently pursuing the Ph.D. degree at the School of Transportation Science and Engineering, Beihang University, Beijing, China. Her main research interests include vehicle platoon control, intelligent transportation systems and robust optimization.



Daxin Tian received his Ph.D degree in computer application technology from Jilin University, Changchun, China, in 2007. He is currently a professor with the School of Transportation Science and Engineering, Beihang University, Beijing, China. His research is focused on intelligent transportation systems, autonomous connected vehicles, swarm intelligent and mobile computing. He was awarded the Changjiang Scholars Program (Young Scholar) of Ministry of Education of China in 2017, the National Science Fund for Distinguished Young Scholars in 2018, and the Distinguished Young Investigator of China Frontiers of Engineering in 2018. He is also a sensor member of the IEEE ans served as the Technical Program Committee member/Chair/Co-Chair for several international conferences including EAI 2018, ICTIS 2019, IEEE ICUS 2019, IEEE HMWC 2020, GRAPH-HOC 2020, etc.



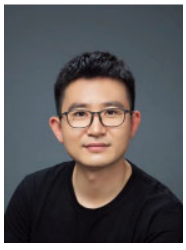
Jianshan Zhou received the B.Sc., M.Sc., and Ph.D. degrees in traffic information engineering and control from Beihang University, Beijing, China, in 2013, 2016, and 2020, respectively. He is an associate professor with the school of transportation science and engineering at Beihang University. From 2017 to 2018, he was a Visiting Research Fellow with the School of Informatics and Engineering, University of Sussex, Brighton, U.K. He was a Postdoctoral Research Fellow supported by the Zhuoyue Program of Beihang University and the National Postdoctoral Program for Innovative Talents from 2020 to 2022. He is or was the Technical Program Session Chair with the IEEE EDGE 2020, the IEEE ICUS 2022, the ICAUS 2022, the TPC member with the IEEE VTC2021-Fall track, and the Youth Editorial Board Member of the Unmanned Systems Technology. His research interests include the modeling and optimization of vehicular communication networks and air–ground cooperative networks, the analysis and control of connected autonomous vehicles, and intelligent transportation systems.



Xuting Duan received the Ph.D degree in Traffic Information Engineering and Control from Beihang University, Beijing, China. He is currently an assistant professor with the School of Transportation Science and Engineering, Beihang University. His current research interests include vehicular ad hoc networks, cooperative vehicle infrastructure system and internet of vehicles.



Dezong Zhao received the B.Eng. and M.S. degrees from Shandong University, Jinan, China, in 2003 and 2006, respectively, and the Ph.D. degree from Tsinghua University, Beijing, China, in 2010, all in Control Science and Engineering. He is a Senior Lecturer in Autonomous Systems with the School of Engineering, University of Glasgow, U.K. Dr Zhao's research interests include connected and autonomous vehicles, machine learning and control engineering. His work has been recognised by being awarded an EPSRC Innovation Fellowship and a Royal Society-Newton Advanced Fellowship in 2018 and 2020, respectively.



Dongpu Cao received the Ph.D. degree from Concordia University, Canada, in 2008. He is the Canada Research Chair in Driver Cognition and Automated Driving, and currently an Associate Professor and Director of Waterloo Cognitive Autonomous Driving (CogDrive) Lab at University of Waterloo, Canada. His current research focuses on driver cognition, automated driving and cognitive autonomous driving. He has contributed more than 200 papers and 3 books. He received the SAE Arch T. Colwell Merit Award in 2012, IEEE VTS 2020 Best Vehicular Electronics Paper Award and three Best Paper Awards from the ASME and IEEE conferences. Prof. Cao serves as Deputy Editor-in-Chief for IET INTELLIGENT TRANSPORT SYSTEMS JOURNAL, and an Associate Editor for IEEE Transactions on Vehicular Technology, IEEE Transactions on intelligent transportation systems, IEEE/ASME Transactions on Mechatronics, IEEE Transactions on Industrial Electronics, IEEE/CAA JOURNAL OF AUTOMATICA SINICA, IEEE Transactions on computational social systems, and ASME Journal of Dynamic Systems, Measurement and Control. He was a Guest Editor for Vehicle System Dynamics, IEEE TRANSACTIONS ON SMC: SYSTEMS and IEEE INTERNET OF THINGS JOURNAL.